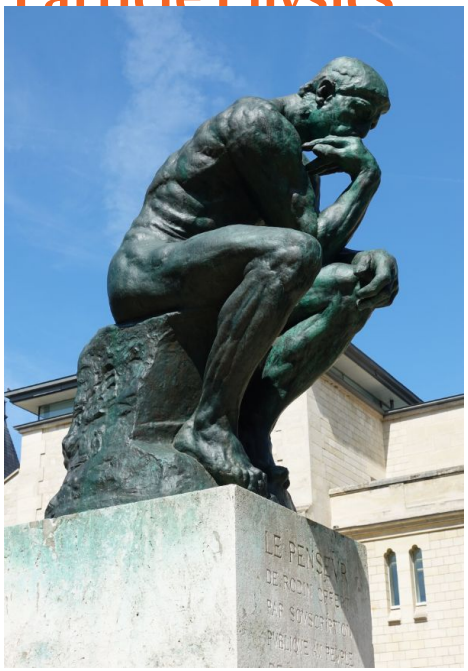


Modern Physics

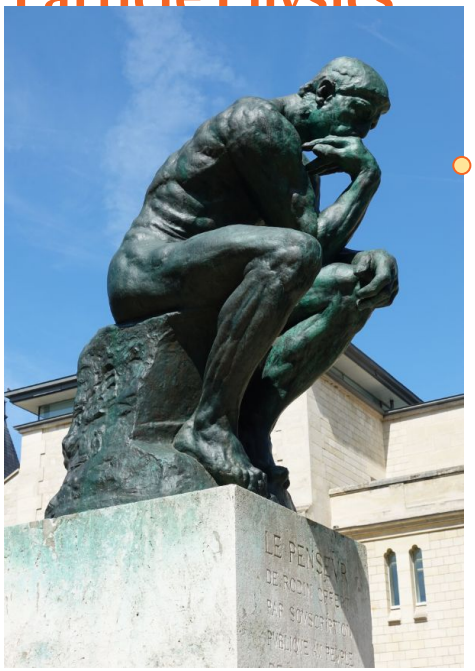
Saurabh Sandilya
Department of Physics,
Indian Institute of Technology Hyderabad.

4th Nov 2024

Quest for fundamentals : Nuclear Physics or Particle Physics



Quest for fundamentals : Nuclear Physics or Particle Physics



What is the world made of? ... What holds it together?..

The Thinker (French: Le Penseur) is a bronze sculpture by **Auguste Rodin**, usually placed on a stone pedestal. The pose is one of deep thought and contemplation, and the statue is often used as an image to represent philosophy.

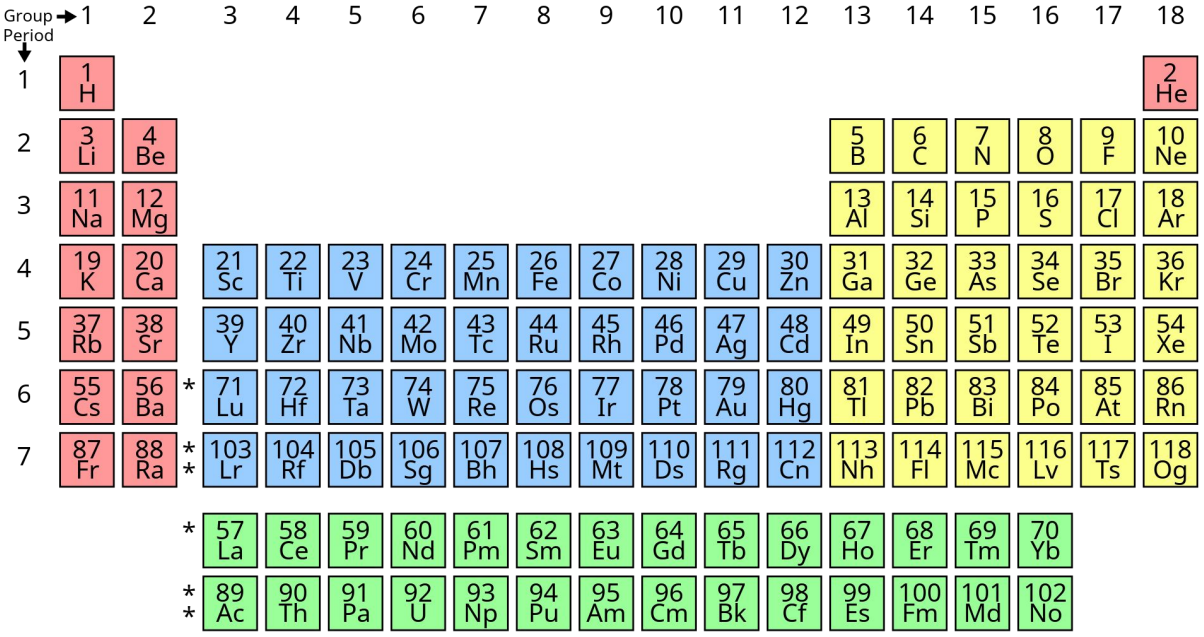
- People have come to realize that the matter of the world is made from a few fundamental building blocks of nature.

- The word "**fundamental**" is key here. By fundamental building blocks we mean objects that are simple and structure-less – not made of anything smaller.
- Even in ancient times, people sought to organize the world around them into fundamental elements, such as earth, air, fire, and water.
- Particle Physics is the quest for finding answer to "What is matter made of?" and "How matter interact with each other?" on the most fundamental level.



Quest for Fundamental

- Dalton's **Atomic Theory** (~1800):
  - All matter is made of very tiny particles called atoms.
  - Atoms are indivisible particles, which cannot be created or destroyed in a chemical reaction.



- Grouped atoms based on their comparable chemical characteristics ⇒ **periodic table**

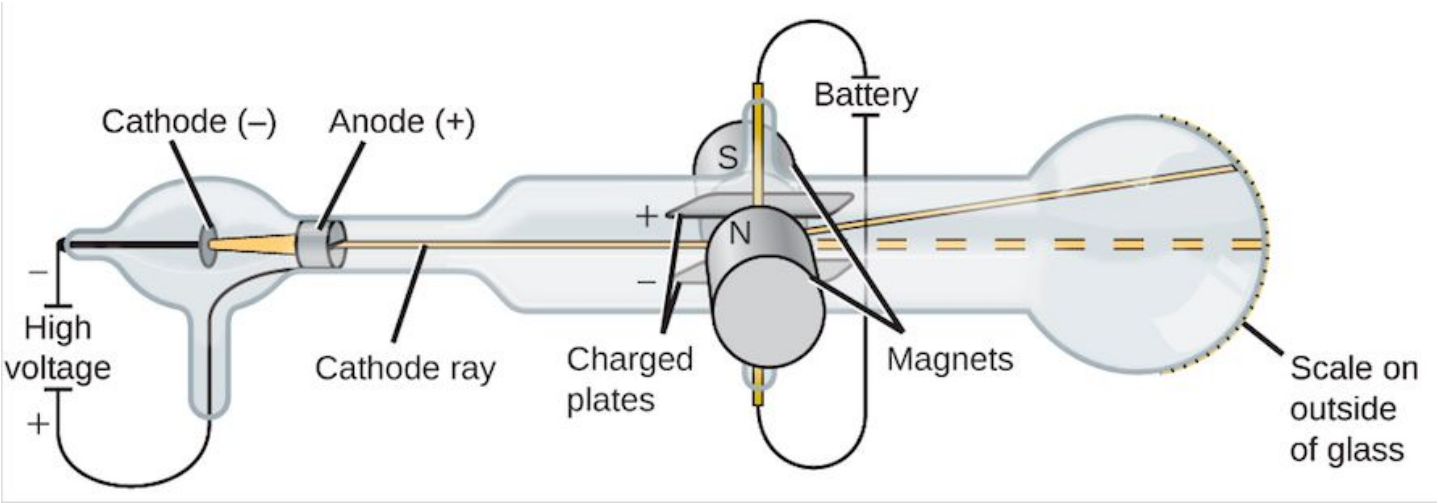
But is the atom fundamental?

Quest for Fundamental : Discovery of electron



"Discovery of electron : J J Thomson"

- Thomson's apparatus was essentially a cathode ray tube.
- Electrons were emitted (via thermionic emission) at the cathode and accelerated towards the anode by a potential difference.



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- Electrons were emitted (via thermionic emission) at the cathode and accelerated towards the anode by a potential difference.
- Thomson calculated the e/m ratio of the electron:

But, How??

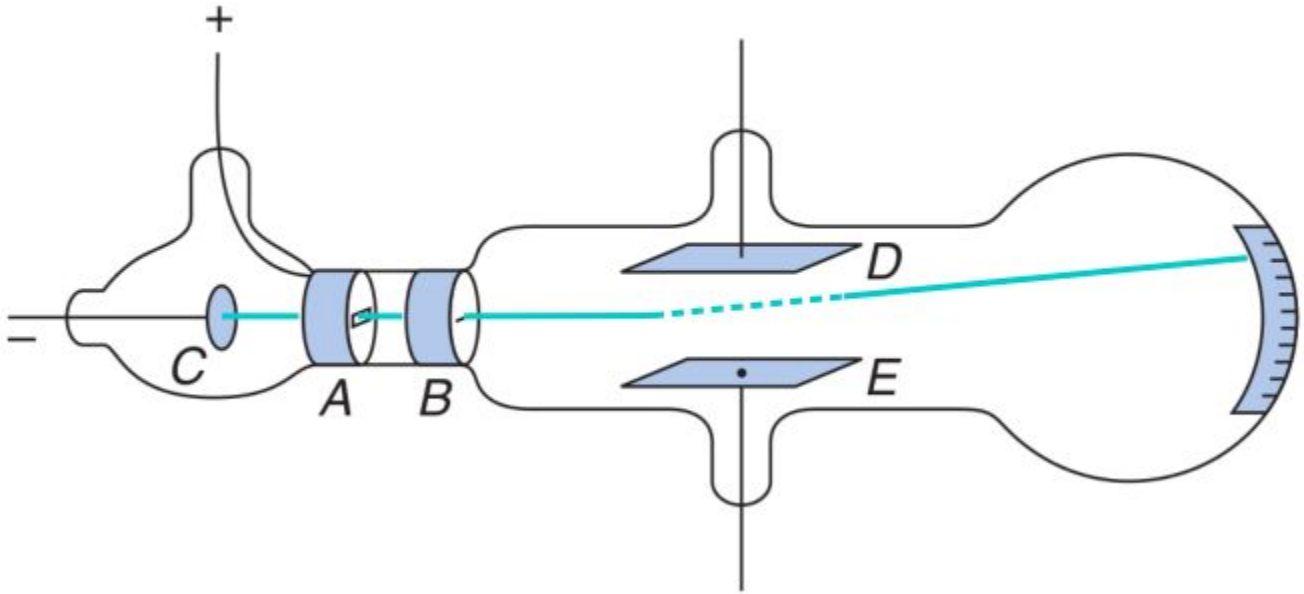


Thomson's Experiment

- In J. J. Thomson experiment, he adjusted perpendicular B and ξ fields so that the electrons were undeflected at the screen.
- This allowed him to determine the speed by equating the magnitudes of the magnetic and electric forces:

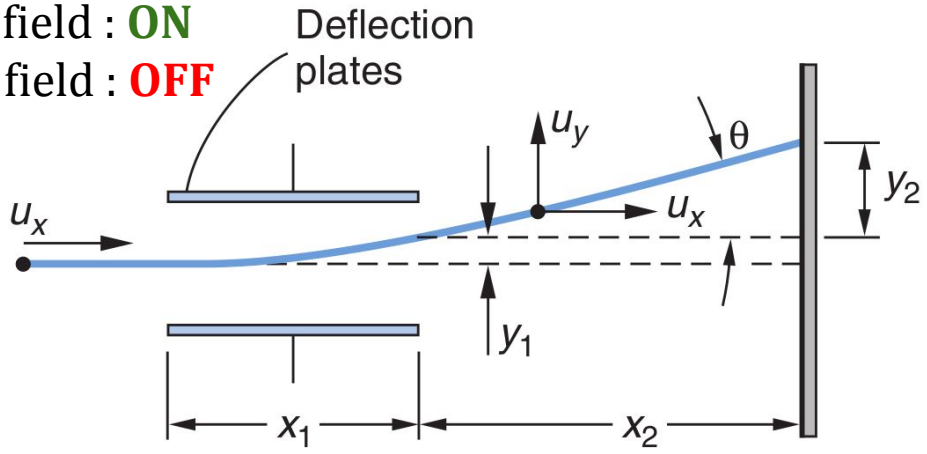
$$quB = q\xi \quad \Rightarrow \quad u = \frac{\xi}{B} \qquad \text{---- (1)}$$

- He then turned off the B field and measured the deflection of the particles on the screen. This deflection is made up of two parts.



Thomson's Experiment

E field : **ON**
B field : **OFF**



- While the particles are between the plates, they undergo a vertical deflection y_1 , given by:

$$y_1 = \frac{1}{2}at_1^2 = \frac{1}{2}\frac{e\xi}{m}\left(\frac{x_1}{u_x}\right)^2$$

t_1 is the same time taken to cross the plate,
x_1 is the plate length.

- After electron leaves the plates, it undergo additional deflection y_2

$$y_2 = u_y t_2 = at_1\left(\frac{x_2}{u_x}\right) = \frac{e\xi}{m}\left(\frac{x_1}{u_x}\right)\left(\frac{x_2}{u_x}\right) = \frac{e\xi}{m}\frac{x_1 x_2}{u_x^2}$$

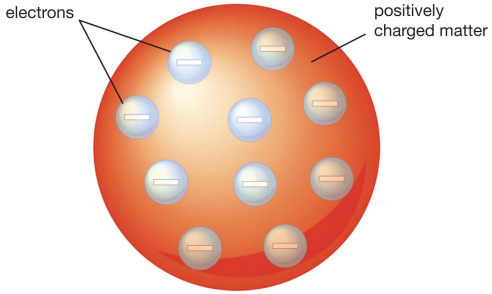
The total deflection ($y_1 + y_2$) can be given by [Note: we have used the relation in eqn (1)]:

$$y_1 + y_2 = \frac{e}{m}\left(\frac{B^2}{\xi}\right)\left(\frac{x_1^2}{2} + x_1 x_2\right)$$

$$\frac{e}{m} = \frac{y_1 + y_2}{\left(\frac{B^2}{\xi}\right)\left(\frac{x_1^2}{2} + x_1 x_2\right)}$$

Quest for Fundamental : Discovery of electron

- Thomson measured e/m ratio by canceling the deflection in trajectory in the presence of electric field by applying a magnetic field of suitable strength.
- This e/m ratio is found to be very high ~1700 times than charged H-atom . Later Millikan, measured the charge on the electron.
- Due to the work of Thomson, Millikan, and others, the charge and mass of the negative, subatomic particles—the electrons—were known.





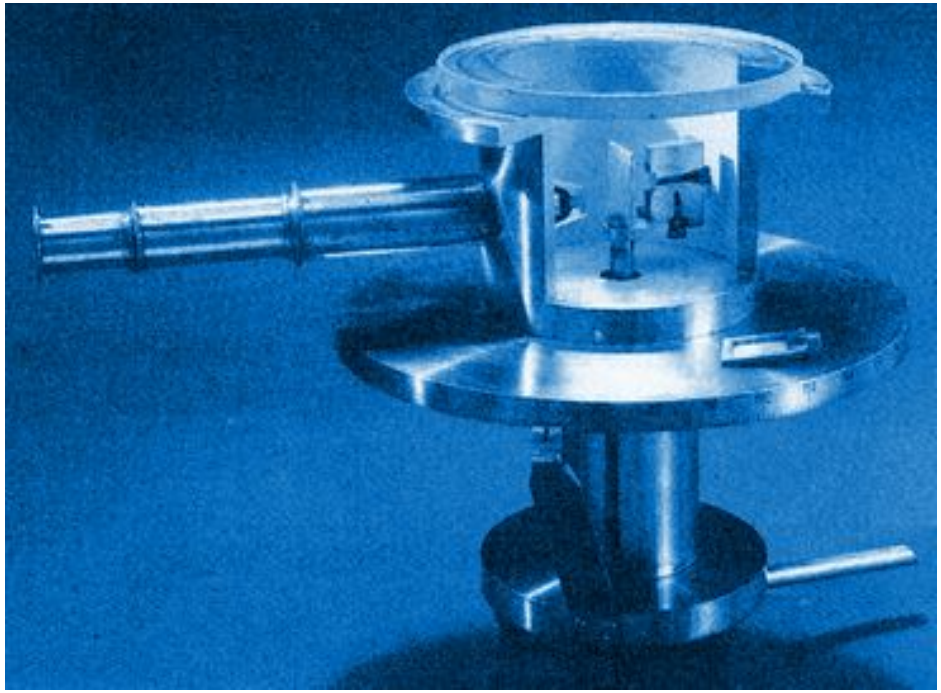
An electron volt (eV) is the amount of kinetic energy gained by a single electron accelerating from rest through an electric potential difference of one volt in vacuum.

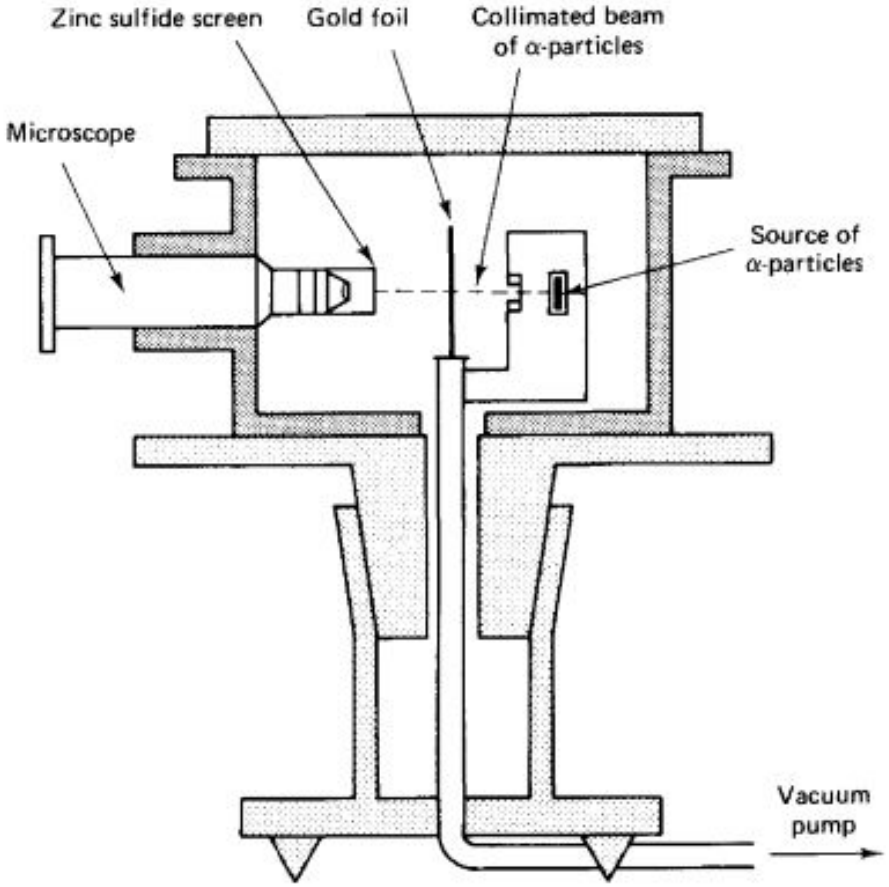
1 eV = ?? J [Hint : Electric potential energy E = qV]

Quest for Fundamental : Discovery of nucleus

"Discovery of the nucleus/proton" – Ernest Rutherford

- The experiments were performed between 1906 and 1913 by Hans Geiger and Ernest Marsden under the direction of Ernest Rutherford at the Physical Laboratories of the University of Manchester.
  - α particles are highly (mono) energetic particles emitted during radioactive decay of certain heavy nuclei.
  - α particles have 2 units of positive charge and mass number 4 ($^4He_2$).

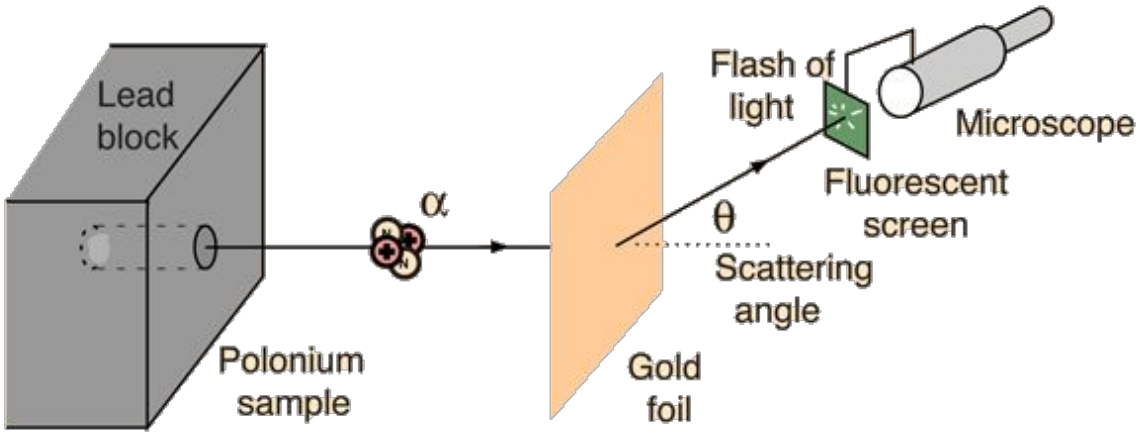




Discovery of nucleus

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  - α particles have 2 units of positive charge and mass number 4 ($^4He_2$).
  - α particles from a radioactive source were allowed to strike a thin gold foil.
  - α particles produce a tiny, but visible flash of light when they strike a fluorescent screen.

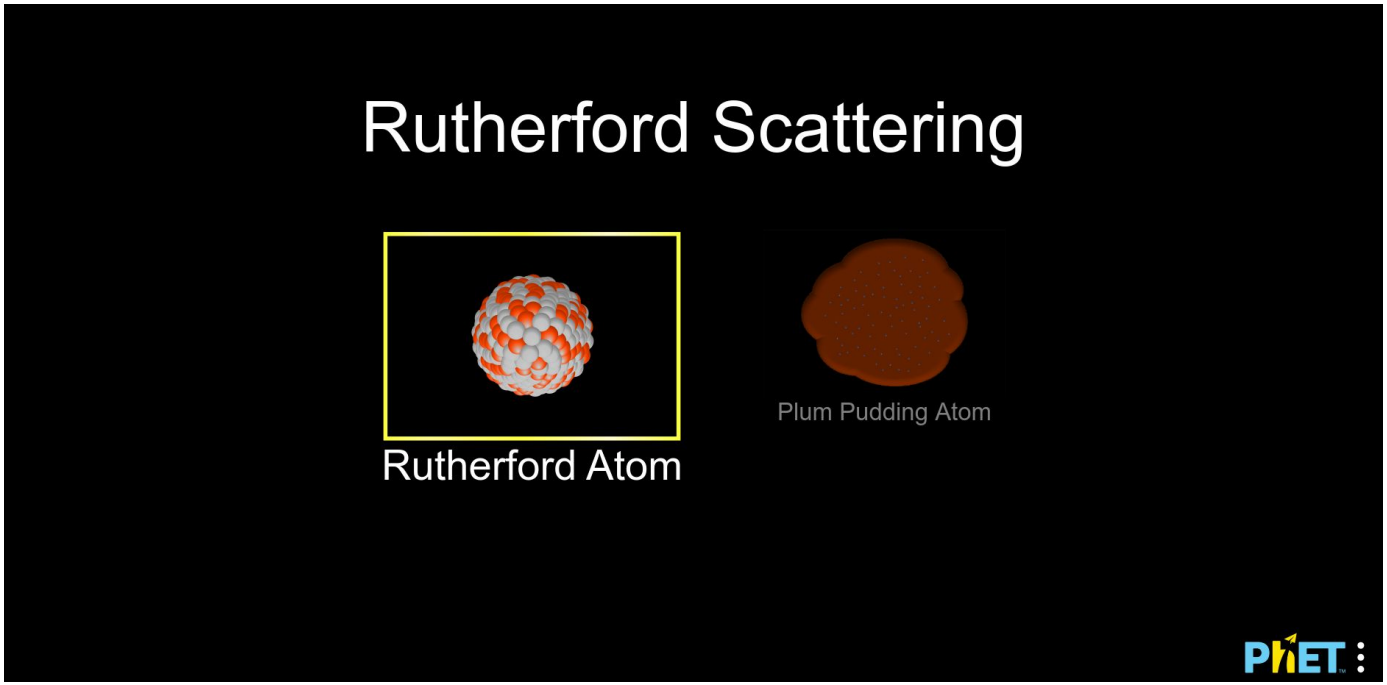


- Rutherford observed that:
  - A majority of α particles are scattered at small angles.
  - A small fraction is found to be scattered at large angles (greater than 90°).
  - Almost one alpha particle among ~15,000 alpha particles bounced back, i.e. deflected nearly 180 degrees.

Discovery of nucleus

"Discovery of the nucleus/proton" – Ernest Rutherford

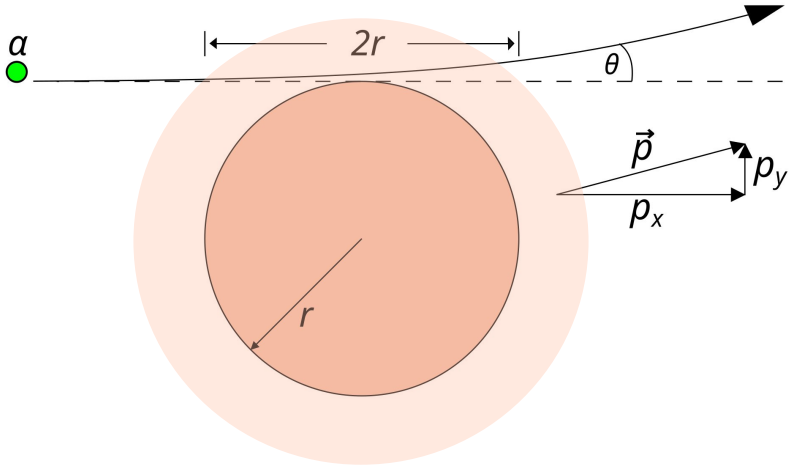
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https://phet.colorado.edu/sims/html/rutherford-scattering/latest/rutherford-scattering_all.html

Rutherford Scattering Formula

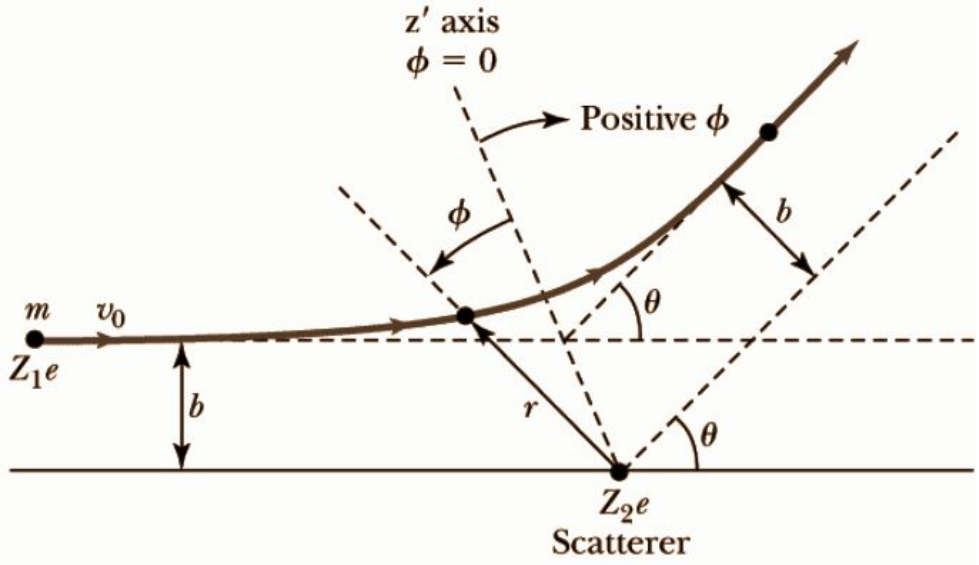
- The Large scattering angles can not be explained on the basis of Thomson's model of atom.
- Force between α particle and scattering atom depends:
  - Inversely on the square of the distance between the α particle and center of the atom.
  - Directly on the positive charge of the scattering atom contained within the sphere of radius r.



- The alpha particle will experience maximum deflection if it just grazes the edge of the positive sphere because that is where the electric field is at its strongest.
  - angle of scattering will be small at large distance from the center of the positive atom.
  - angle of scattering will be small at small distance from the center of the positive atom (positive charge contained is small).

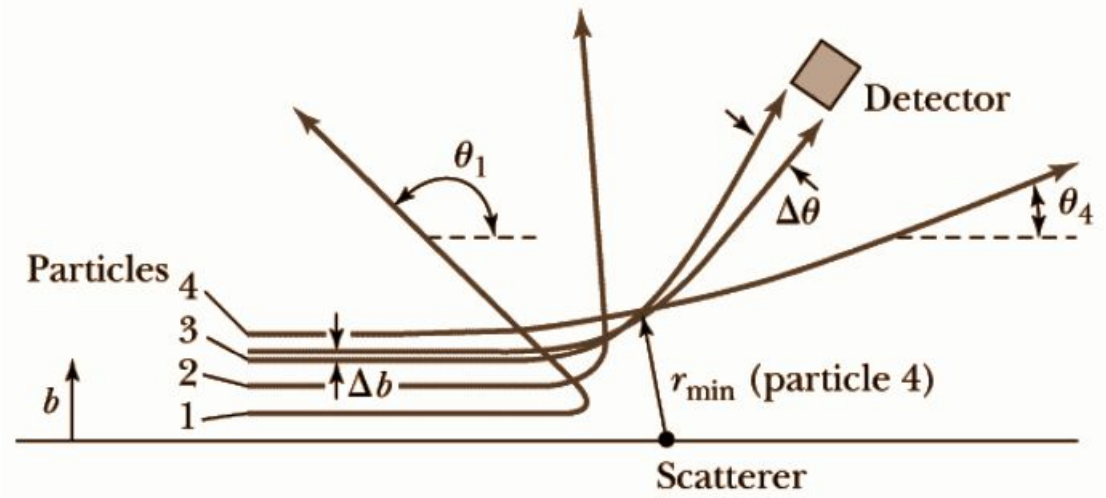
Rutherford Scattering Formula

- To explain large scattering angle Rutherford proposed entire positive charge of the atom (and almost entire mass) to be contained within a very small sphere near center : NUCLEUS.
- When an α particle passes close by to the nucleus it experiences a strong electrostatic force due to entire positive charge of the atom, and hence scattered by a relatively large angle.
- The force experienced by the α particle is central => the trajectories are hyperbolic.



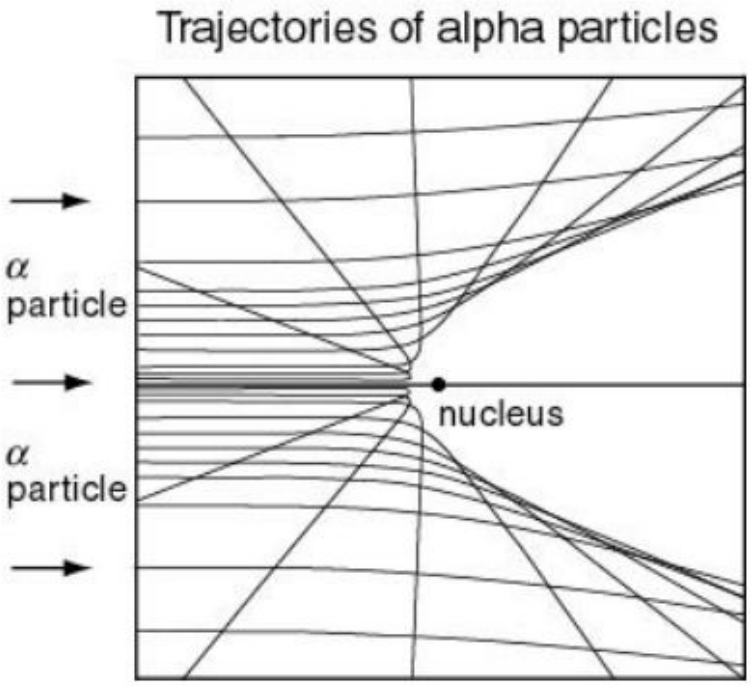
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  => the trajectories depends on the impact parameter.



Rutherford Scattering Formula

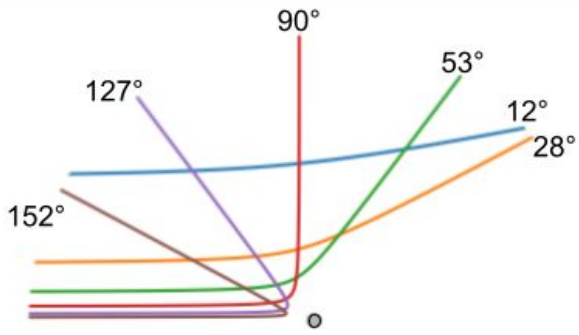
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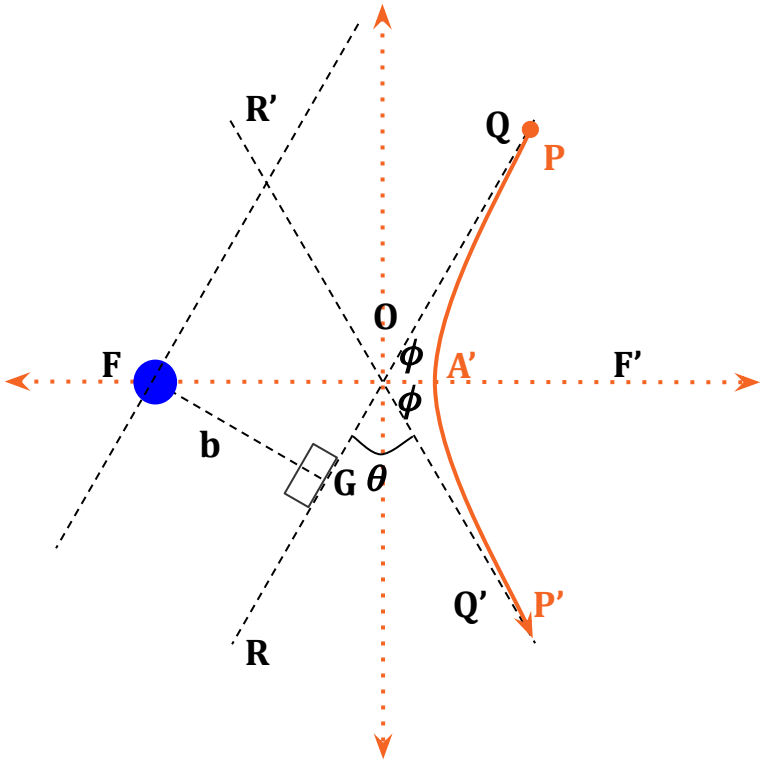
Hyperbolic trajectories of alpha particles scattering from a Gold nucleus (modern radius shown as gray circle) as described in Rutherford's 1911 paper

Let's derive Rutherford formula :
Impact parameter vs. scattering angle

$$b = \frac{Z Z' e^2}{4\pi\epsilon_0 M v^2} \cdot \cot\left(\frac{\theta}{2}\right)$$



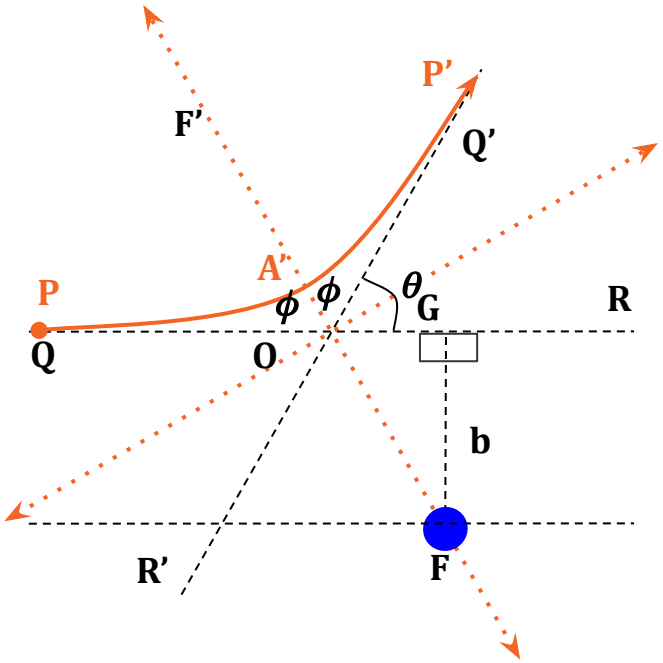
Impact Parameter *vs.* scattering angle



- PA'P' is the α-particle scattering (hyperbolic) path.
- The scattering nucleus is at F (one of its Foci) F' is another foci.
- Entire charge of the nuclei (+Ze) is on the point F.
- mass of the α-particle = M
- Charge of the α-particle = +z'e
- Mass of the nuclei is considered infinitely heavy as compared to the α-particle (assumption).
- Initially, the entire energy of α-particle is kinetic:

$$E_\alpha = \frac{1}{2} M v^2$$

Impact Parameter *vs.* scattering angle



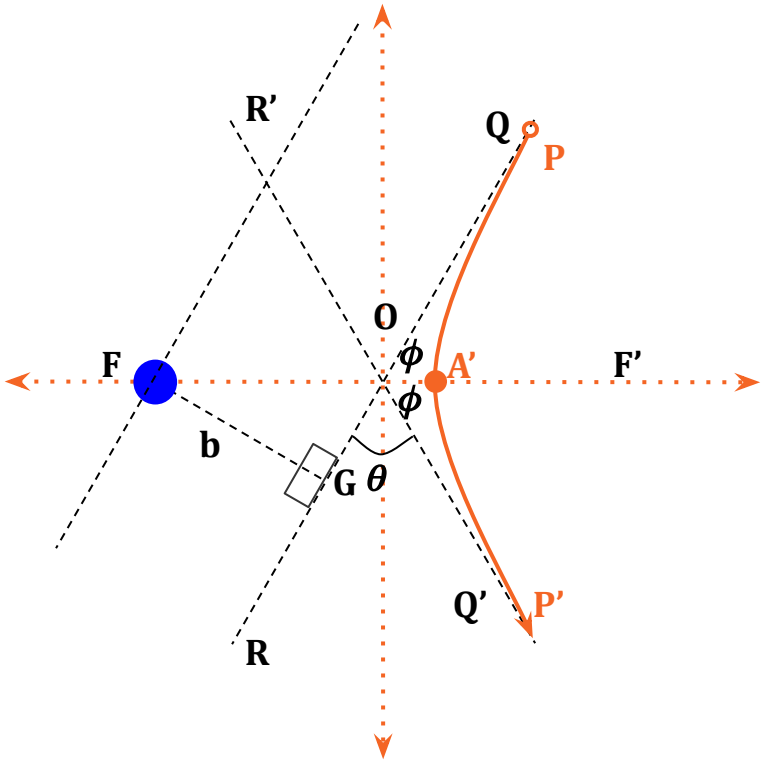
(rotated diagram)

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- Impact Parameter : Perpendicular to the asymptote of the hyperbola (FG) = b
- Scattering Angle: angle between the two asymptotes of the hyperbola ($\angle$Q'OR) = θ

Impact Parameter *vs.* scattering angle



- The two asymptotes are equally inclined at an angle ϕ to the axis FF' (joining the two foci) of the hyperbola.
- When α-particle reaches minimum distance (FA' = q) the electrostatic repulsion is maximum.
- At A'
  - Potential Energy:

$$V = \frac{Z z' e^2}{4\pi\epsilon_0 q}$$

  - Kinetic Energy:

$$\frac{1}{2} M v_m^2$$

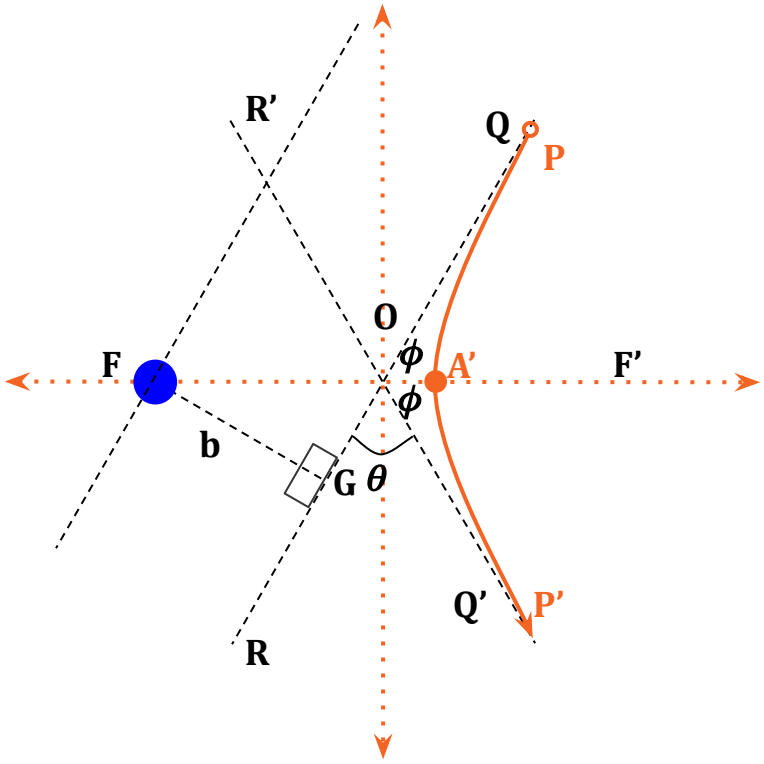
- Equating the initial and final energy:

$$\frac{1}{2} M v^2 = \frac{1}{2} M v_m^2 + \frac{Z z' e^2}{4\pi\epsilon_0 q}$$

- Re-arranging:

$$\frac{v_m^2}{v^2} = 1 - \frac{Z z' e^2}{2\pi\epsilon_0 M v^2 q} \quad \text{--- (1)}$$

Impact Parameter *vs.* scattering angle



- At P (Initial condition)
  - α-particle momentum = Mv
  - α-particle angular momentum = Mv.(FG) = Mv.b
- At A' (at minimum distance to the nuclie)
  - α-particle momentum = Mv_m
  - α-particle angular momentum = Mv_m.(FA') = Mv_m.q
- The force acting on the α-particle is central ⇨ Angular momentum must be conserved.

  $Mv.b = Mv_m.q$ --- (2)
- From the properties of hyperbola:

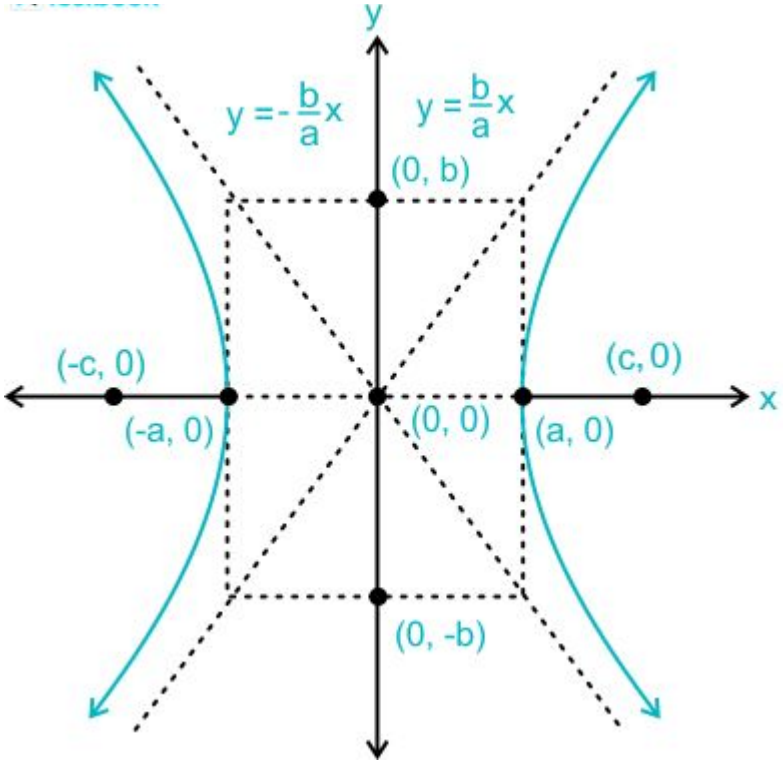
  OF = OF' = εOA' (eccentricity, where ε > 1)
  - q = FA' = OF + OA'
  - **$q = OF(1+ 1/\varepsilon)$ --- (3)**

- And, from the theory of conics:
  - **$1/\varepsilon = \cos\phi$ --- (4)**

Assignment : Conic Section properties

Consider the hyperbola that is centered at the origin and horizontally oriented, then the equation of the hyperbola is:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



Let $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be the hyperbola

It has asymptotes $y = \pm\frac{b}{a}x$

Angle between the asymptotes $= 2\tan^{-1}1(\frac{b}{a}) = \theta$

$\Rightarrow \tan\frac{\theta}{2} = \pm\frac{b}{a}$

$\Rightarrow \sec^2\frac{\theta}{2} = 1 + \tan^2\frac{\theta}{2} = 1 + \frac{b^2}{a^2}$

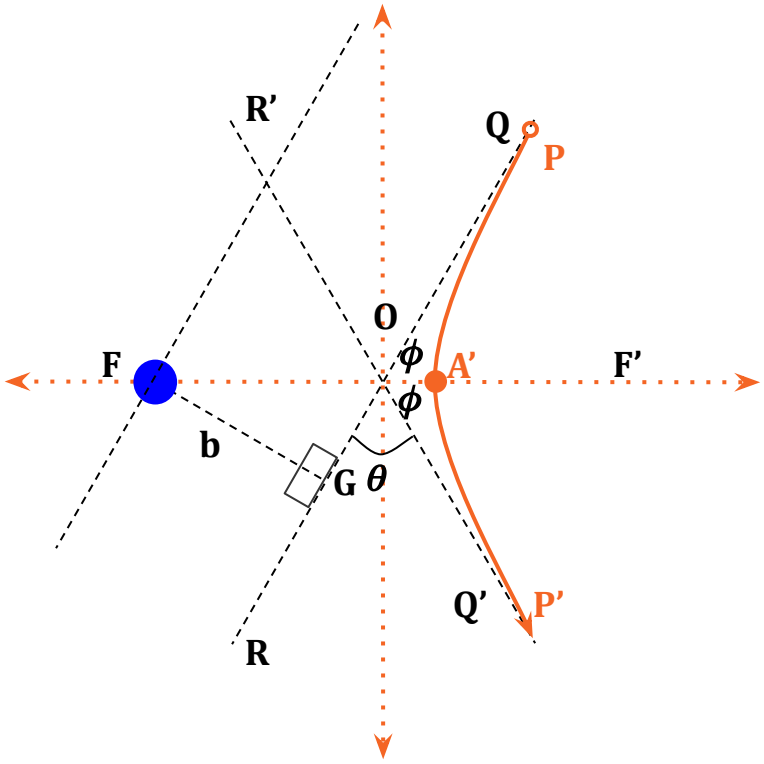
$\Rightarrow \sec^2\frac{\theta}{2} = \sqrt{1 + \frac{b^2}{a^2}}$

$\Rightarrow \sec^2\frac{\theta}{2} = e^2$

$\Rightarrow \cos\frac{\theta}{2} = \frac{1}{e}$



Impact Parameter *vs.* scattering angle



- From equation (3) and (4)

  $q = OF(1+\cos\phi)$ --- (5)

- And,

  $\sin\phi = FG/OF$ --- (6)

  $\Rightarrow \quad OF = FG/\sin\phi$

- From equation (5) and (6)

$$q = \frac{FG(1+\cos\phi)}{\sin\phi} \quad \{FG = b\}$$

$$q = \frac{b(1+\cos\phi)}{\sin\phi} \quad \text{--- (7)}$$

$$\frac{b}{q} = \frac{\sin\phi}{(1+\cos\phi)} \quad \text{--- (8)}$$

From Equation (2)

$$\frac{b}{q} = \frac{v_m}{v} = \frac{\sin\phi}{1+\cos\phi}$$

$$\Rightarrow \quad 1 - \frac{Zz'e^2}{2\pi\epsilon_0 M v^2 q} = \frac{v_m^2}{v^2} = \frac{\sin^2\phi}{(1+\cos\phi)^2}$$

$$\Rightarrow \quad 1 - \frac{Zz'e^2}{2\pi\epsilon_0 M v^2} \cdot \frac{\sin\phi}{b(1+\cos\phi)} = \frac{1-\cos^2\phi}{(1+\cos\phi)^2}$$

↑ value of q substituted from eq^n (7)

$$\Rightarrow \quad 1 - \frac{Zz'e^2}{2\pi\epsilon_0 M v^2} \cdot \frac{\sin\phi}{b(1+\cos\phi)} = \frac{1-\cos\phi}{1+\cos\phi}$$

Multiplying by $(1+\cos\phi)$ both side L.H.S. and R.H.S.

$$\Rightarrow \quad 1+\cos\phi - \frac{Zz'e^2 \sin\phi}{2\pi\epsilon_0 M v^2 b} = 1-\cos\phi$$

Impact Parameter *vs.* scattering angle

$$\Rightarrow \quad 1+\cos\phi - \frac{Zz'e^2 \sin\phi}{2\pi\epsilon_0 M v^2 b} = 1-\cos\phi$$

$$\Rightarrow \quad \frac{Zz'e^2 \sin\phi}{2\pi\epsilon_0 M v^2 b} = 2\cos\phi$$

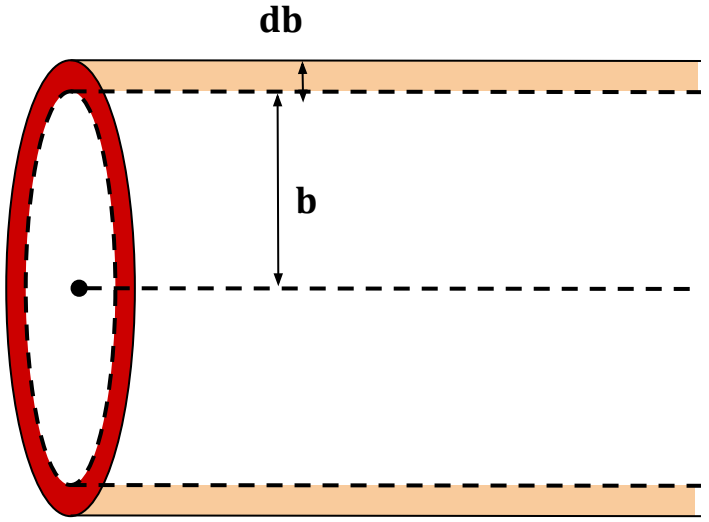
$$\Rightarrow \quad b = \frac{Zz'e^2}{4\pi\epsilon_0 M v^2} \tan\phi \qquad \text{---} (10)$$

$$\left[\phi + \frac{\theta}{2} = \frac{\pi}{2}\right]$$

$$\Rightarrow \quad \boxed{b = \frac{Zz'e^2}{4\pi\epsilon_0 M v^2} \cot\frac{\theta}{2}} \qquad \text{---} (11)$$

Rutherford's scattering formula

- Let **N** be the flux of the α-particle incident per unit area per unit time.
- α-particles having impact parameter between b and b+db will be along the co-axial cylinder of radius b and b+db.
- Area (red ring) = $2\pi b.db$
- Number of α-particles (per unit time): $dN = N.2\pi b.db$



$$\Rightarrow \frac{db}{d\theta} = \frac{Zz'e^2}{4\pi\epsilon_0 M v^2} \frac{d(\cot\theta/2)}{d\theta}$$

$$\Rightarrow \frac{db}{d\theta} = \frac{Zz'e^2}{4\pi\epsilon_0 M v^2} \cdot \left[-\frac{1}{2}\cdot \operatorname{cosec}^2\theta/2\right]$$

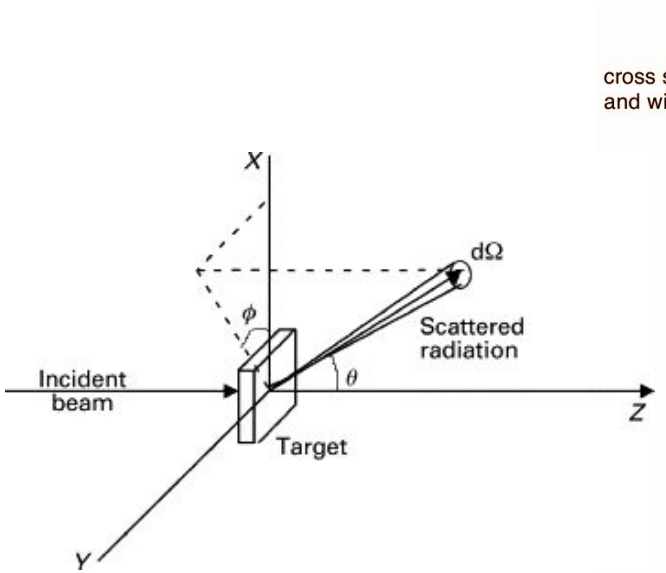
$$\Rightarrow dN = 2\pi b \cdot N \cdot \left[-\frac{1}{2}\cdot\frac{Zz'e^2}{4\pi\epsilon_0 M v^2}\cdot \operatorname{cosec}^2\theta/2\right] d\theta$$

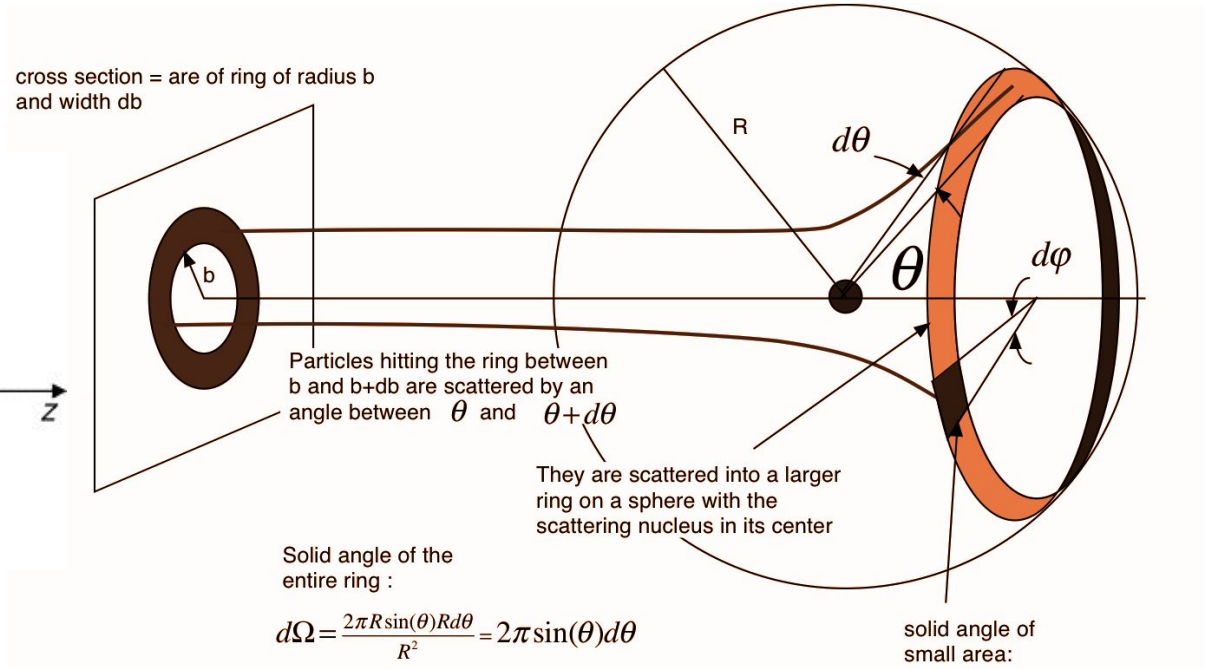
→ substituting value of b from eq^n (11)

$$\Rightarrow dN = -\pi N \left(\frac{Zz'e^2}{4\pi\epsilon_0 M v^2}\right)^2 \operatorname{cosec}^2\theta/2 \cdot \cot\theta/2 \cdot d\theta \quad \text{--- (12)}$$

Rutherford's scattering formula

- The α-particles within the annular region (db) are scattered in a solid angle dΩ.





$$\Rightarrow \quad d\Omega = 2\pi \sin\theta\, d\theta = 2\pi\left(2\cdot \sin\theta/2 \cdot \cos\theta/2\right) d\theta$$

$$\Rightarrow \quad d\Omega = 4\pi \sin\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta \qquad \text{--- (13)}$$

Rutherford's scattering formula

- The number of α-particles scattered per unit solid angle is (from equation 12 and equation 13)

$$\frac{dN}{d\Omega} = \frac{(-)N}{4} \cdot \left(\frac{Z z' e^2}{4\pi\epsilon_0 M v^2}\right)^2 \operatorname{cosec}^4 \theta/2$$

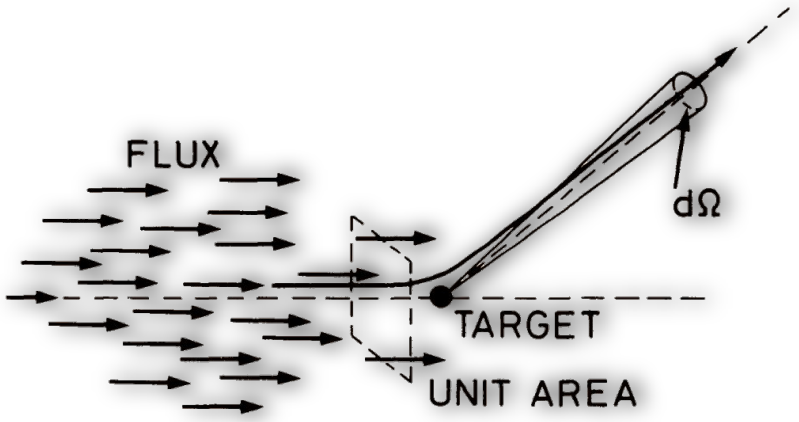
- The (−) sign simply indicate that as the impact parameter 'b' increases, the scattering angle decreases.
- The α-particles has z' = 2

$$\frac{dN}{d\Omega} = N \cdot \left(\frac{Z e^2}{4\pi\epsilon_0 M v^2}\right)^2 \operatorname{cosec}^4\left(\frac{\theta}{2}\right)$$

Rutherford's scattering formula

Cross-section

- beam flux = N incident particles per unit area per unit time.
- Let's now examine the rate at which particles (α) are scattered into a specific solid angle (dΩ) over a fixed period.
- The stochastic distribution of impact parameters leads to statistical fluctuations in the number of scattered particles observed within finite measurement windows.



- Averaging across measurements yields a stable value for dN/dΩ (average scattered particles per unit time and solid angle).

$$\frac{dN}{d\Omega} = N \cdot \left(\frac{Z e^2}{4\pi\epsilon_0 M v^2}\right)^2 \operatorname{cosec}^4\left(\frac{\vartheta}{2}\right)$$

- Our next focus is on the differential scattering cross-section, represented by dσ/dΩ, which signifies the number of particles (dN) scattered into a small solid angle dΩ per unit time and per unit incident flux N.

$$\frac{d\sigma}{d\Omega} = \left(\frac{1}{N}\frac{dN}{d\Omega}\right) = \left(\frac{Z e^2}{4\pi\epsilon_0 M v^2}\right)^2 \operatorname{cosec}^4\left(\frac{\vartheta}{2}\right)$$

What is the unit of dσ/dΩ?

Experimental Verification

- Experimental verification of Rutherford's scattering formula Geiger and Marsden.
- From the Rutherford's scattering formula we can conclude:

$$N_S \propto \frac{1}{\sin^4 \theta/2}$$

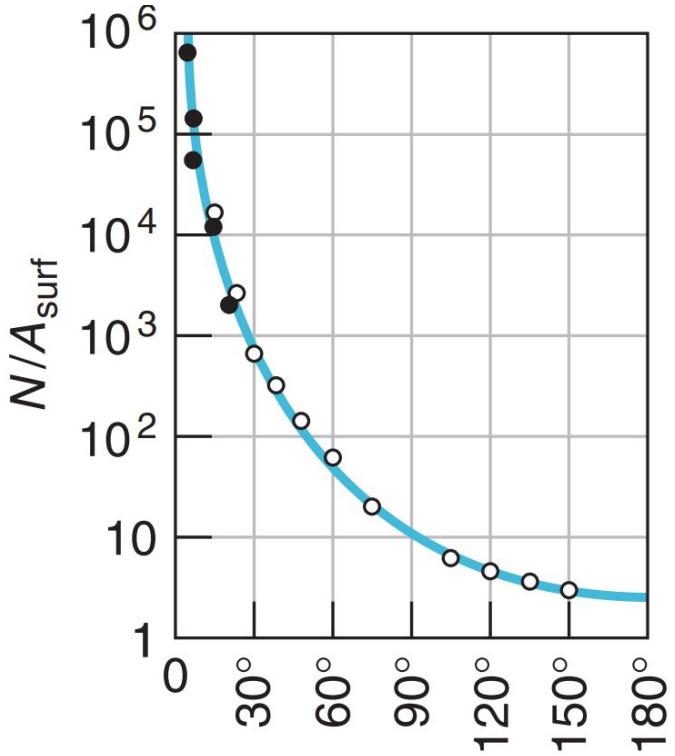
$$N_S \propto t \quad \text{(thickness of the foil)}$$

$$N_S \propto 1/E^2 \quad \{E = \tfrac{1}{2} M v^2 \text{ K.E. of the } \alpha\text{-particle}\}$$

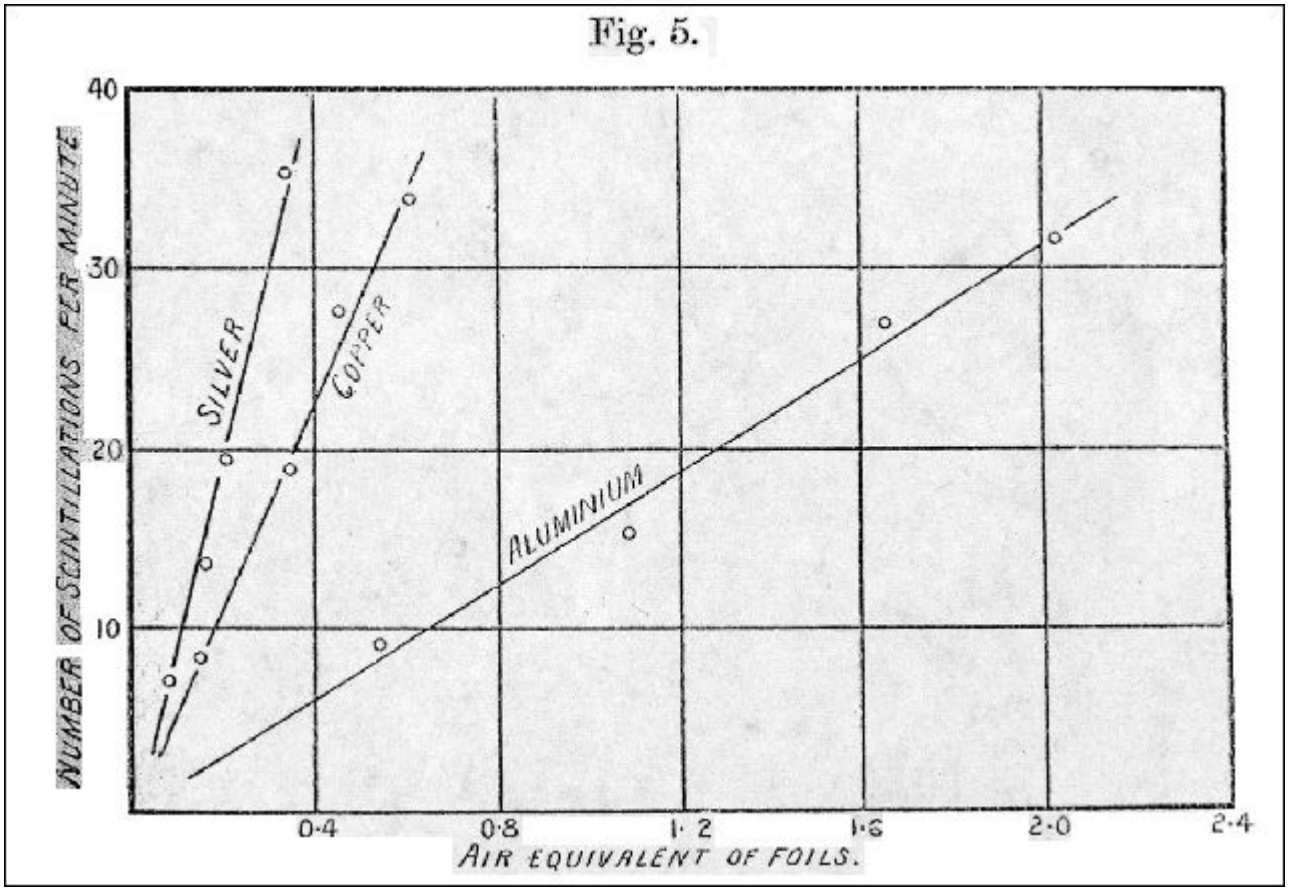
$$N_S \propto Z^2$$

As a function of scattering angle

Angle f	Scintillations per minute.				$\frac{1}{\sin^4 f}$	N x ($\sin^4 f$ / 2)
	Without foil.	With foil.	Corrected for effect without foil.	Corrected for decay, N.		
150	0.2	4.95	4.75	6.95	1.15	6.0
135	2.6	8.3	5.7	8.35	1.38	6.1
120	3.8	10.3	6.5	9.5	1.79	5.3
105	0.6	10.6	10.0	14.6	2.53	5.8
75	0.0	28.6	28.6	41.9	7.25	5.8
60	0.3	69.2	68.9	101	16.0	6.3

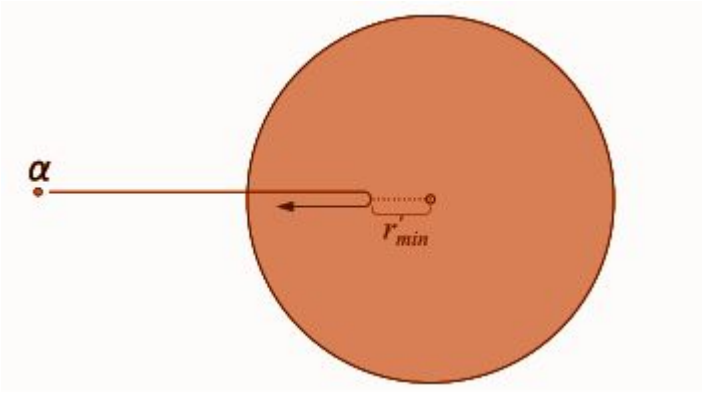


Thickness and Z



"Estimating" Maximum Nuclear size from Rutherford scattering

- Examining a head-on collision between an alpha particle and an atom.
- This collision helps determine the minimum distance between the nucleus and the alpha particle.
- In a head-on collision, the alpha particle's entire kinetic energy is converted to potential energy.
- This energy conversion brings the alpha particle to a complete stop before reversing its direction.



- Where the particle stops, a distance r_{min} the potential energy matches the original kinetic energy:

$$\frac{1}{2}mv^2 = k\frac{q_1 q_2}{r_{\min}} \qquad \text{where} \qquad k = \frac{1}{4\pi\epsilon_0}$$

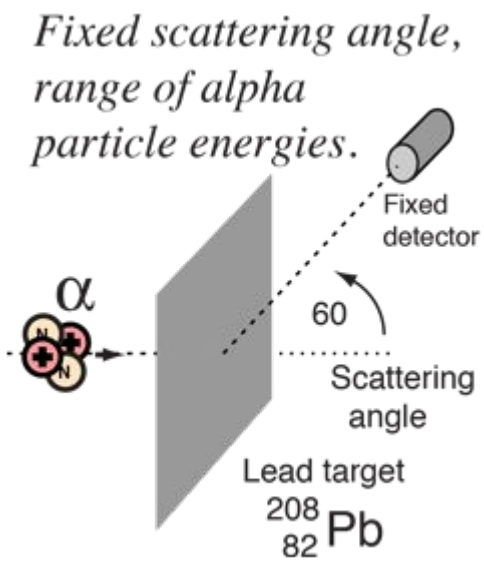
- Rearranging: $$r_{\min} = k\frac{2q_1 q_2}{mv^2}$$

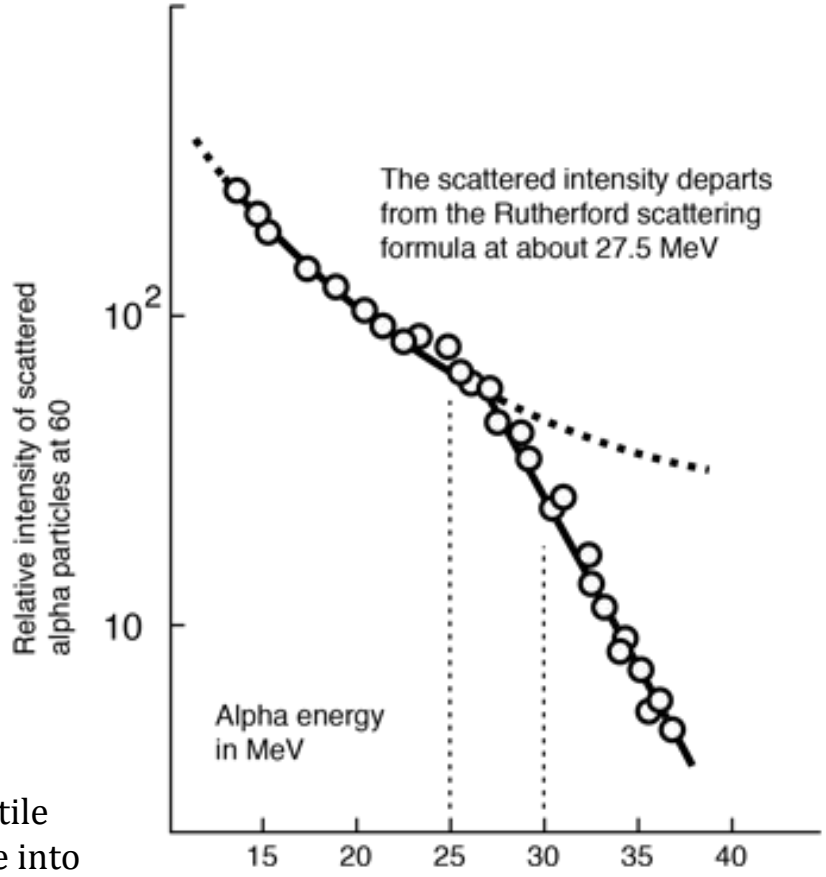
- The distance from the alpha particle to the center of the nucleus (r_{min}) at this point is an upper limit for the nuclear radius. Substituting these in gives the value of about 2.7×10^{-14} m, or 27 fm. (The true radius is about 7.3 fm.)

Source : wikipedia, S.N.Ghosal

Departure From Rutherford Formula

- Scattering of high energy alpha particles at a fixed angle can be used to estimate nuclear radius.





- With high enough alpha energies, however, the projectile punches in close enough to the nuclear center to come into range of the nuclear strong force and the distribution of scattered alphas departs from the Rutherford formula.

- Considering the data above for scattering from ^{208}Pb, one can use the point of departure from the Rutherford scattering formula to estimate the nuclear radius.

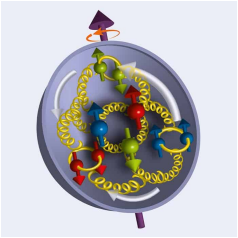
Nucleus and discovery of proton

- In 1886, Eugen Goldstein identified positively charged particles, called canal rays, produced from gases.
- Unlike negatively charged electrons, these particles varied in charge-to-mass ratio by gas type.
- Hydrogen ion had the highest charge-to-mass ratio among these particles.
- After Rutherford's discovery of the atomic nucleus in 1911, Moseley experimentally verified that an element's atomic number corresponds to its nuclear charge using X-ray spectra.
- Moseley experimentally verified this hypothesis in 1913 using X-ray spectra.
- Rutherford's experiments (1917), revealed the presence of the hydrogen nucleus within other nuclei, marking the discovery of protons.
- These experiments were initiated by the observation of scintillations on a ZnS screen caused by particles ejected from air when struck by alpha particles.

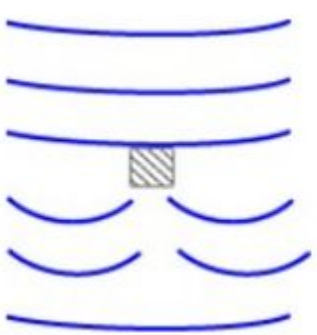
$$^{14}N + \alpha \rightarrow {}^{17}O + p$$

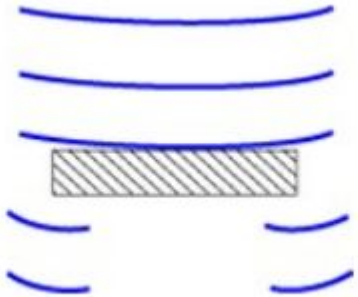
- The distance of these scintillations exceeded the range of alpha particles but matched the range of hydrogen atoms, suggesting the presence of protons.

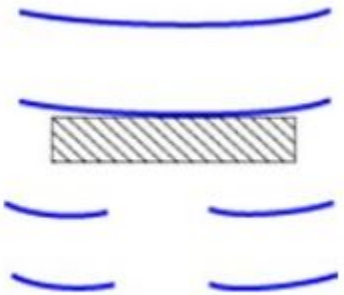
- Proton still puzzles :
  - Proton spin: https://physicsworld.com/a/the-spin-of-a-proton/
  - Proton decays?
  - Proton radius : spectroscopic method vs scattering method. (https://physicsworld.com/a/has-the-proton-radius-puzzle-been-solved-at-last/)



Measurement of Nuclear Radius : Need of High Energy Beam







Negligible Diffraction: When light with a very large wavelength encounters obstructions that are significantly smaller than its wavelength, the light essentially passes through them without significant interaction.

For eg: Ocean wave and pebbles, Radio wave and very small objects like pen, chalk

For **pronounced diffraction**, the size of an obstacle or aperture should be comparable to the wavelength of light

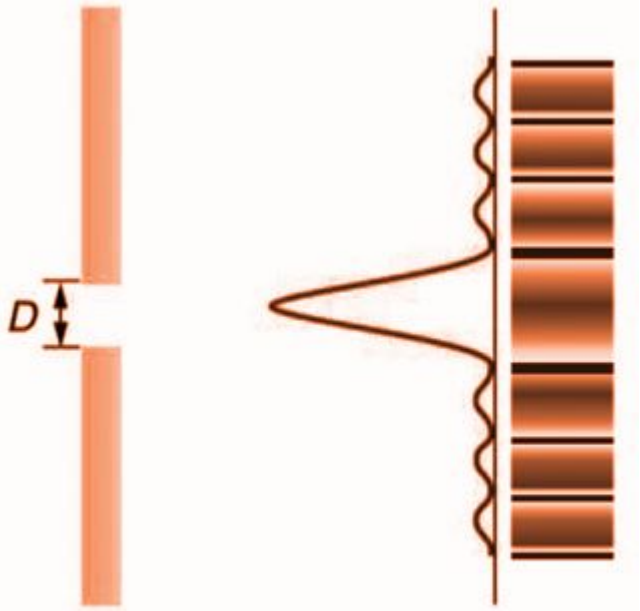
Negligible Diffraction: Diffraction is practically nonexistent due to the extremely large size of the obstruction relative to the wavelength.

Single Slit Diffraction

- If we have a single slit of width "D".
- Monochromatic, coherent light of wavelength λ falls on it.
- Intensity of distribution of the light on a screen at a large distance can be given by:

$$I = I_0 \frac{\sin^2\beta}{\beta^2}$$

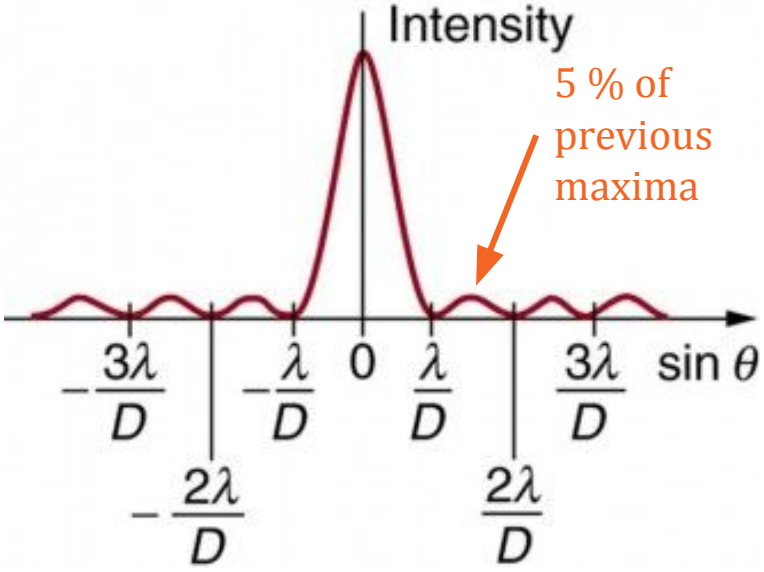
where, $\beta = \frac{\pi}{\lambda} D \sin\theta$



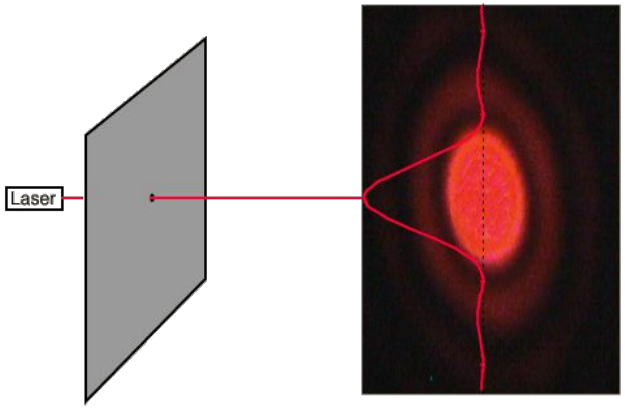
- Intensity minima will occur at:

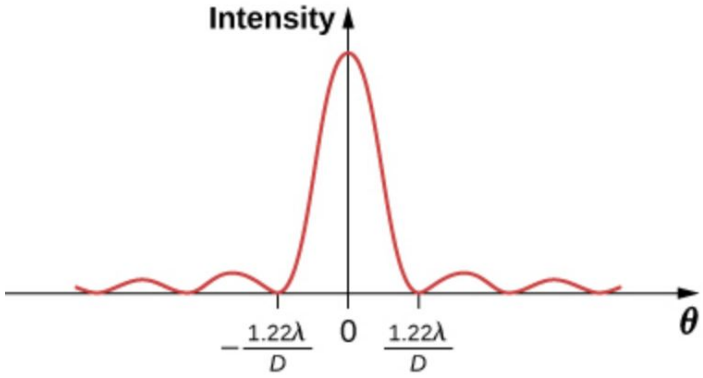
$$D.\sin\theta = \pm\lambda, \pm 2\lambda, \ldots$$

- At θ = 0, intensity is maximum
- Also note that maximum intensity is also falling.



Circular Aperture Diffraction





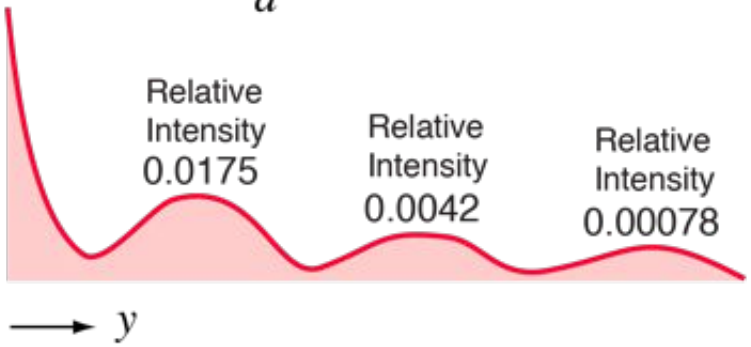
Circular aperture

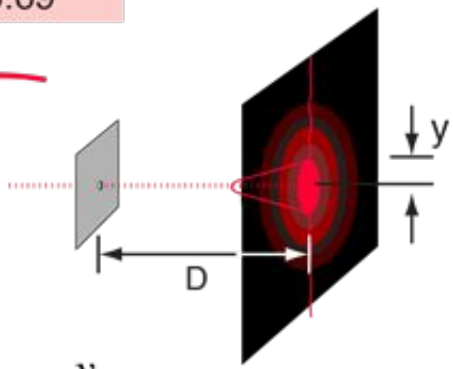
$$\sin\theta = \frac{m\lambda}{d}$$

d = aperture diameter

	m values for: Minima	Maxima
1	1.220	1.635
2	2.233	2.679
3	3.238	3.69

$$y \approx D\frac{m\lambda}{d}$$ for maxima and minima





$$\frac{y}{D} = \tan\theta \approx \sin\theta \approx \theta$$

for small angles θ

For circular aperture, first minima occurs at

$$\sin\theta = \frac{1.22\lambda}{D}$$

De-Broglie Wavelength of electron beam

- All particles exhibit wave-like characteristics. The de Broglie wavelength quantifies the scale at which these wave properties become significant.

$$\lambda = h/p$$

- For a high energy electrons:

$$E = (p^2c^2 + m_e^2c^4)^{1/2}$$

- As the K.E. is very high as compared to the rest mass, we can neglect the rest mass:

$$E = (p^2c^2 + \cancel{m_e^2c^4}^{0})^{1/2} \; ; \; E \simeq pc$$

- As the K.E. is very high as compared to the rest mass, we can neglect the rest mass:

$$\lambda \simeq \frac{hc}{E}$$

$$hc = 1.98 \times 10^{-25} \text{ J.m} \qquad hc = 1240 \text{ MeV.fm}$$

- Calculate the de-Broglie wavelength of electron beam of Energy 190 MeV and 420 MeV respectively.

De-Broglie Wavelength of electron beam

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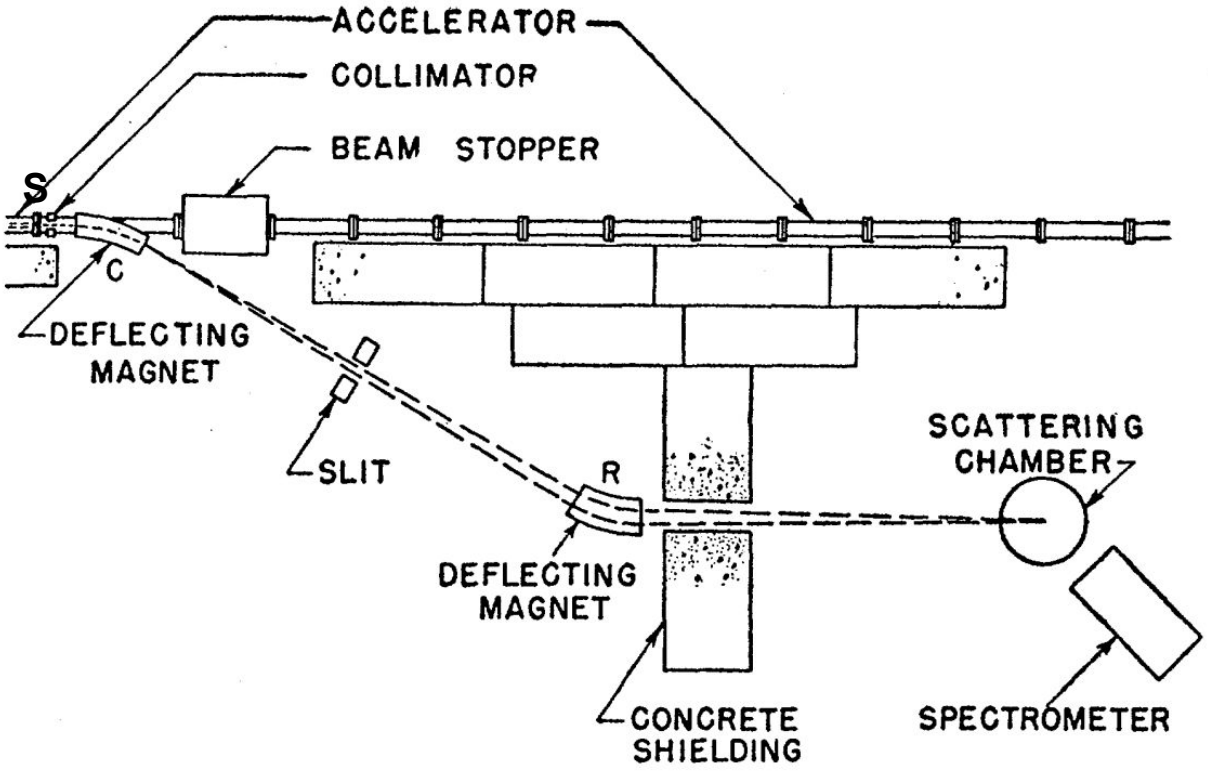
$$\lambda \simeq \frac{hc}{E}$$

$hc = 1.98 \times 10^{-25}$ J.m $hc = 1240$ MeV.fm

- Calculate the de-Broglie wavelength of electron beam of Energy 190 MeV and 420 MeV respectively.

Answer: 6.53 fm for 190 MeV and 2.95 fm for 420 MeV

190 MeV apparatus - R. Hofstadter et al. (SLAC)



- **Halfway station**: Starting at approximately the halfway point along the length of the linear accelerator.
- A shielded vault encloses the medium energy (up to 190 MeV) spectrometer and its associated equipment.
- After passing through a brass collimator, a group of accelerated electrons of nearly uniform energy, is deflected (and dispersed) by magnet C.

- The magnet R straightens the beam again to its original direction and refocuses it to a point approximately 9 feet from the end of the magnet R.
- The principal idea of the double deflection is to obtain a gamma (bremsstrahlung) free beam of electrons.

Robert Hofstadter, Rev. Mod. Phys. 28, 214 (1956)
R. Hofstadter, H. R. Fechter, and J. A. McIntyre, Phys. Rev. 92, 978 (1953)

190 MeV apparatus - R. Hofstadter et al. (SLAC)

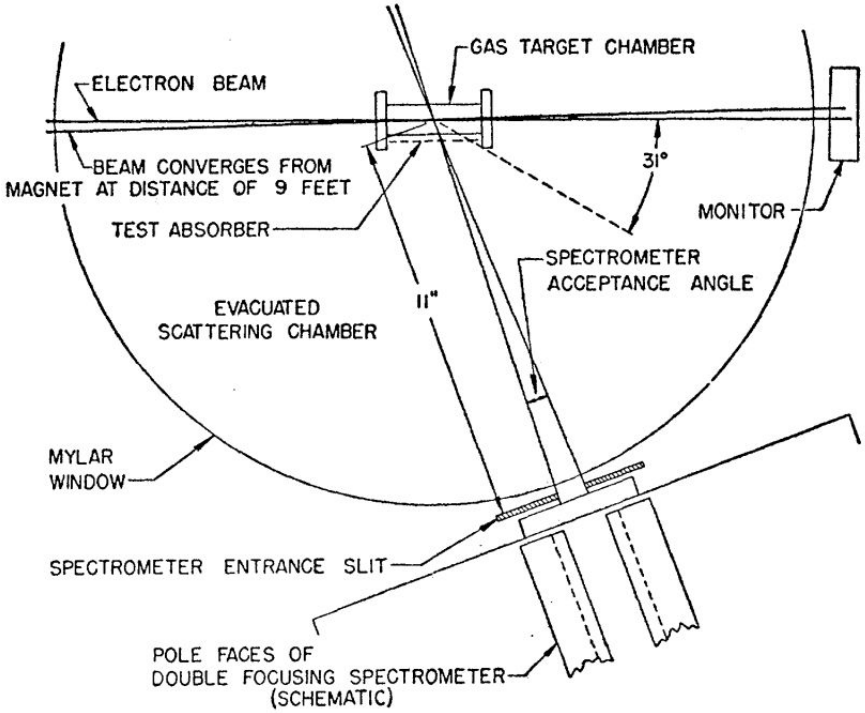


FIG. 17. Schematic diagram of scattering geometry employed with the gas target chamber.

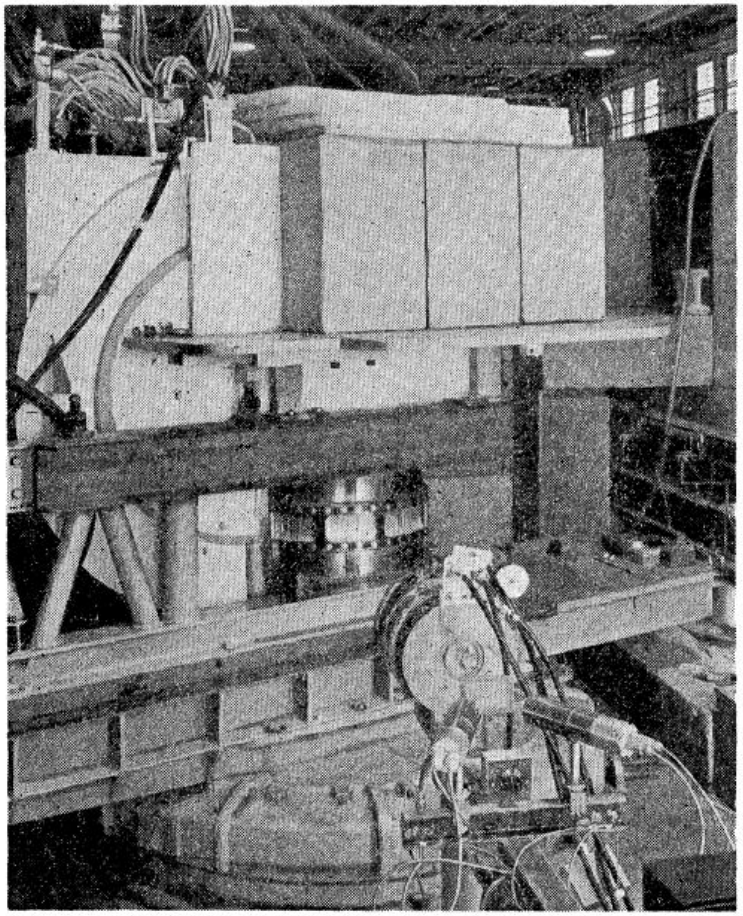


FIG. 15. The semicircular 190-Mev spectrometer, to the left, is shown on the gun mount. The upper platform carries the lead and paraffin shielding that encloses the Čerenkov counter. The brass scattring chamber is shown below with the thin window encircling it. Ion chamber monitors appear in the foreground.

Robert Hofstadter, Rev. Mod. Phys. 28, 214 (1956)
R. Hofstadter, H. R. Fechter, and J. A. McIntyre, Phys. Rev. 92, 978 (1953)

Electron scattering with 420 MeV beam : Results

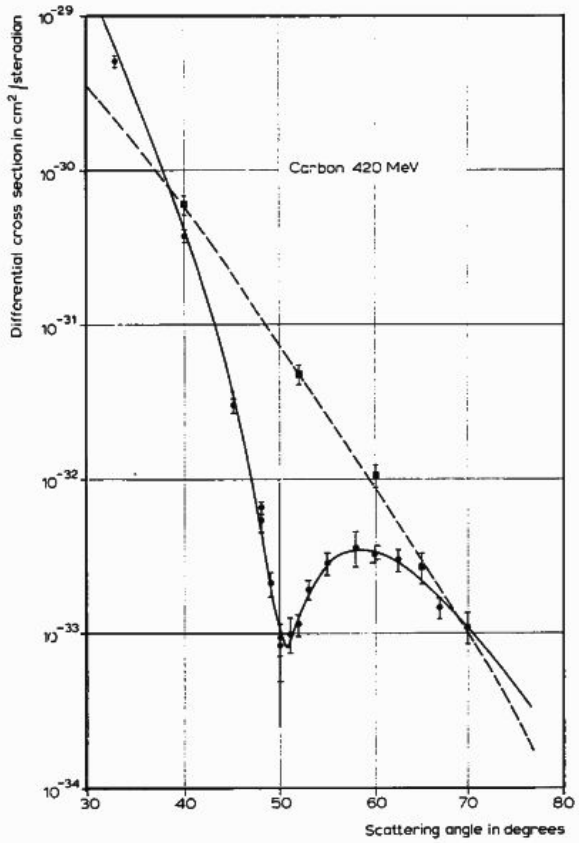


Fig. 5. This figure shows the elastic and inelastic curves corresponding to the scattering of 420-MeV electrons by ^{12}C. The *solid circles*, representing experimental points, show the elastic-scattering behavior while the *solid squares* show the inelastic-scattering curve for the 4.43-MeV level in carbon. The *solid line* through the elastic data shows the type of fit that can be calculated by phase-shift theory for the model of carbon shown in Fig. 8.

Hofstadter, R.: The Electron-Scattering Method and Its Application to the Structure of Nuclei and Nucleons, Nobel Lecture, 1961, https://www.nobelprize.org/uploads/2018/06/hofstadter-lecture.pdf

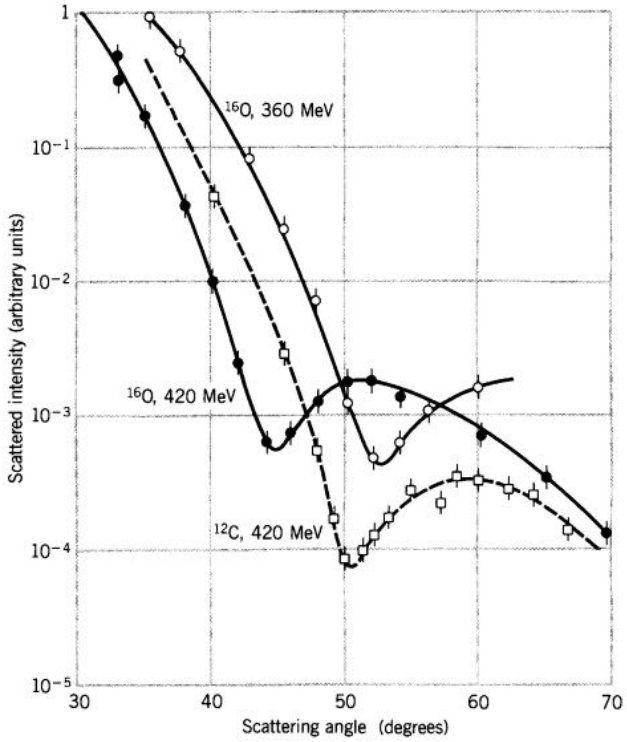


Figure 3.1 Electron scattering from ^{16}O and ^{12}C. The shape of the cross section is somewhat similar to that of diffraction patterns obtained with light waves. The data come from early experiments at the Stanford Linear Accelerator Center (H. F. Ehrenberg et al., *Phys. Rev.* **113**, 666 (1959)).

Introduction to Nuclear Physics, K. S. Krane

Electron scattering with 420 MeV beam : Results

- In general scattering intensity decreases with scattering angle
  $\Rightarrow$ like Rutherford scattering $\propto \text{cosec}^4(\theta/2)$
  (with relativistic corrections applied)
- First minima should occurs at :

$$\sin\theta = \frac{1.22\lambda}{D}$$

- Let's try to estimate the nuclear size (^{16}O) from the results of electron scattering experiments.

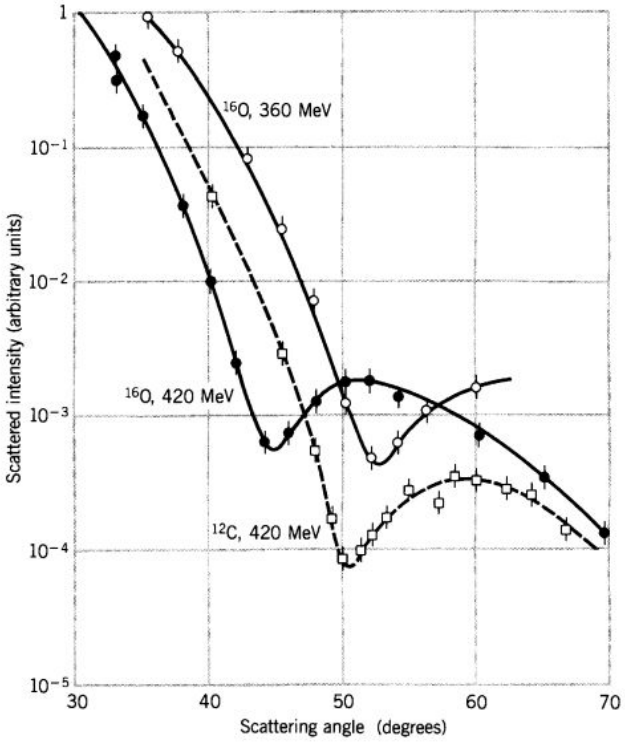


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Introduction to Nuclear Physics, K. S. Krane

Estimate the nuclear radius with scattering data

- First minima occurs at : $\sin\theta = \frac{1.22\lambda}{D}$

$$\Rightarrow D = \frac{1.22\lambda}{\sin\theta}$$

$$\Rightarrow D = \frac{1.22\lambda}{\sin(43^\circ)} \simeq \frac{1.22\lambda}{\sin(45^\circ)}$$

$$\Rightarrow D = \frac{1.22 \times 2.95}{1/\sqrt{2}} = 1.22 \times 2.95 \times 1.41 \text{ fm}$$

$$\Rightarrow D = 5.07 \text{ fm}$$

$$\Rightarrow r = D/2 = 5.07 \text{ fm}/2 = 2.54 \text{ fm}$$

- **Question:** Similarly, estimate the nuclear radius (^{12}C) from the results of electron scattering experiments. (First minima occurs at $\theta = 50^\circ$ in 420 MeV electron beam energy)

Estimate the nuclear radius with scattering data

- First minima occurs at : $\sin\theta = \frac{1.22\lambda}{D}$

$$\Rightarrow D = \frac{1.22\lambda}{\sin\theta}$$

$$\Rightarrow D = \frac{1.22\lambda}{\sin(43^\circ)} \simeq \frac{1.22\lambda}{\sin(45^\circ)}$$

$$\Rightarrow D = \frac{1.22 \times 2.95}{1/\sqrt{2}} = 1.22 \times 2.95 \times 1.41 \text{ fm}$$

$$\Rightarrow D = 5.07 \text{ fm}$$

$$\Rightarrow r = D/2 = 5.07 \text{ fm}/2 = 2.54 \text{ fm}$$

- **Question:** Similarly, estimate the nuclear radius (^{12}C) from the results of electron scattering experiments. (First minima occurs at $\theta = 50^\circ$ in 420 MeV electron beam energy)
- Answer: 2.35 fm

What about other Minima?

- We observed only first minima, what about other minima.
- Let's calculate the second minima for ^{16}O:

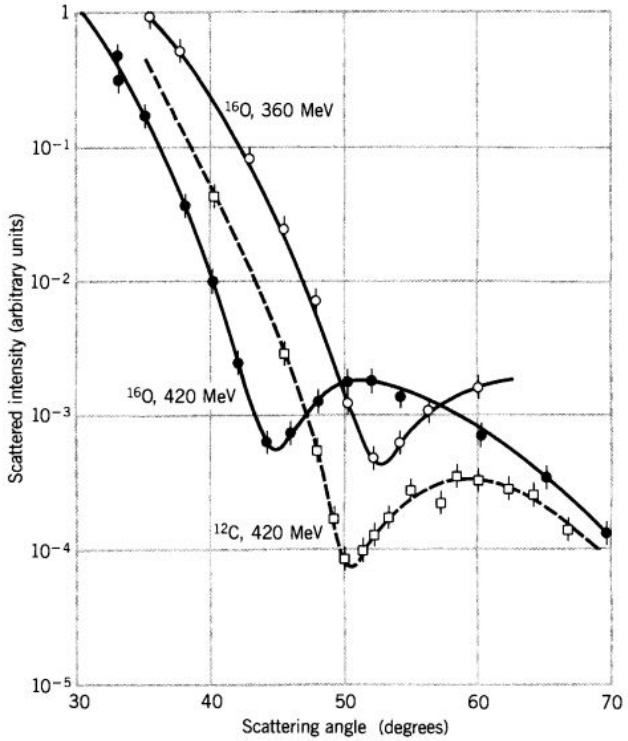


Figure 3.1 Electron scattering from ^{16}O and ^{12}C. The shape of the cross section is somewhat similar to that of diffraction patterns obtained with light waves. The data come from early experiments at the Stanford Linear Accelerator Center (H. F. Ehrenberg et al., *Phys. Rev.* **113**, 666 (1959)).

What about other Minima?

- We observed only first minima, what about other minima.
- Let's calculate the second minima for ^{16}O:

- We estimated the nuclear radius of ^{16}O = 2.35 fm.
- Nuclear diameter D = 4.7 fm
- Second minima should occur at:

$$\sin\theta = \frac{2.33\lambda}{D}$$

- λ = 2.95 fm for 420 MeV electron beam.

$$\sin\theta = \frac{2.33 \times 2.95}{4.7}$$

$$\sin\theta \neq 1.46$$

$\sin\theta > 1$ not possible

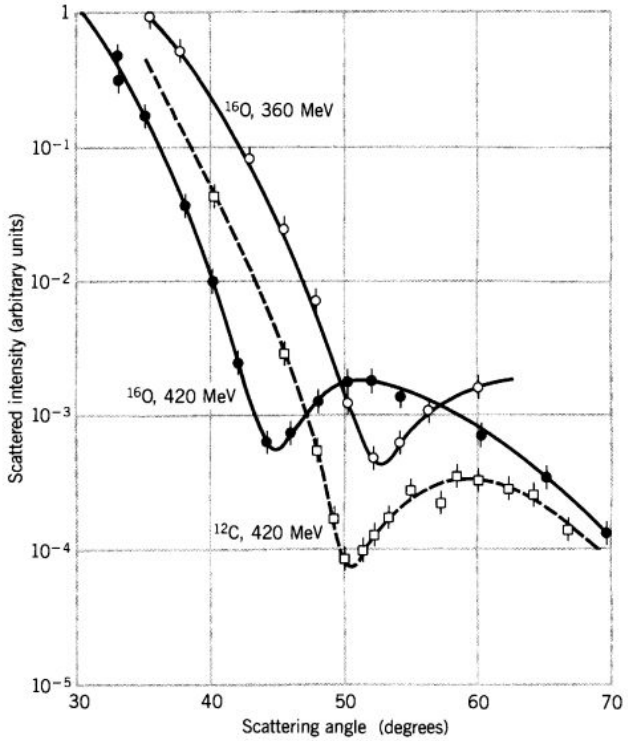


Figure 3.1 Electron scattering from ^{16}O and ^{12}C. The shape of the cross section is somewhat similar to that of diffraction patterns obtained with light waves. The data come from early experiments at the Stanford Linear Accelerator Center (H. F. Ehrenberg et al., *Phys. Rev.* **113**, 666 (1959)).

What about other Minima?

- Other (than first) minima do occur : for example electron scattering with ^{208}Pb nuclei.

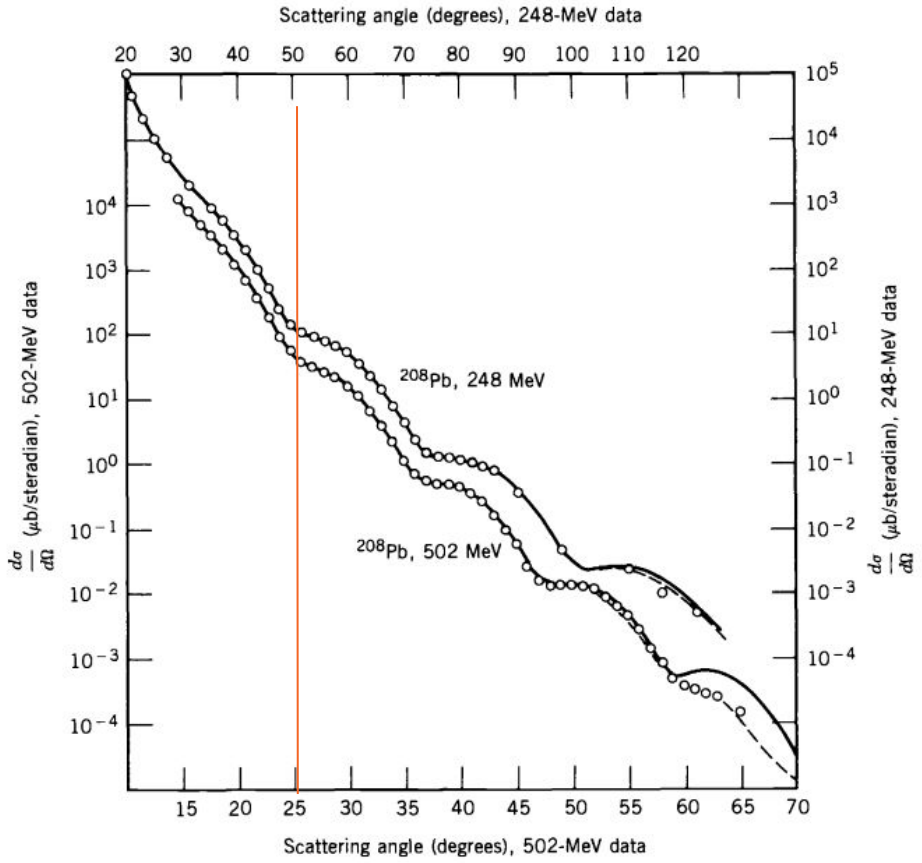


Figure 3.2 Elastic scattering of electrons from ^{208}Pb. Note the different vertical and horizontal scales for the two energies. This also shows diffractionlike behavior, but lacks sharp minima. (J. Heisenberg et al., *Phys. Rev. Lett.* **23**, 1402 (1969).)



Caveat:

- The nuclear radii estimates are good but needs to be corrected for the case of 3D as we considered the scattering from the 2D disk.
- The minima doesn't fall to zero like diffraction pattern ⇒ nucleus doesn't have a sharp boundary

Nuclear charge distribution (proton density)

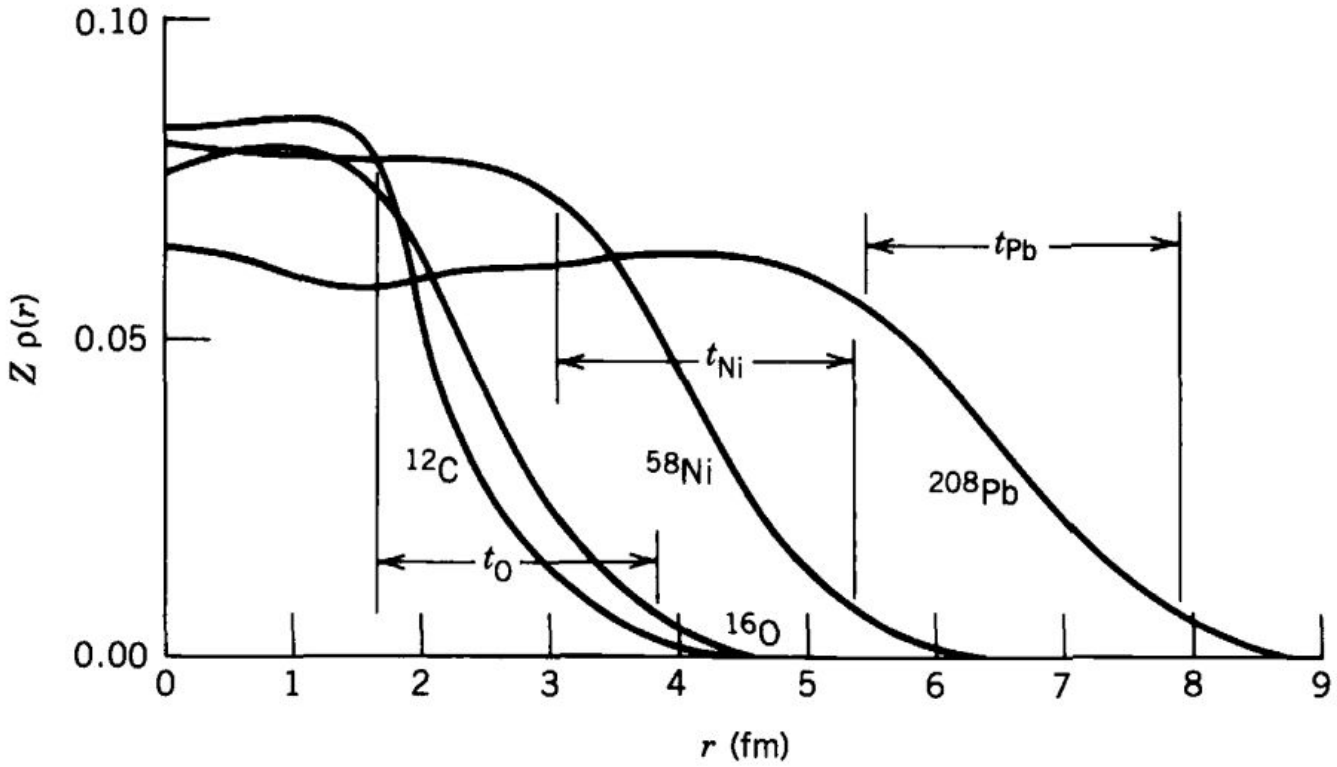
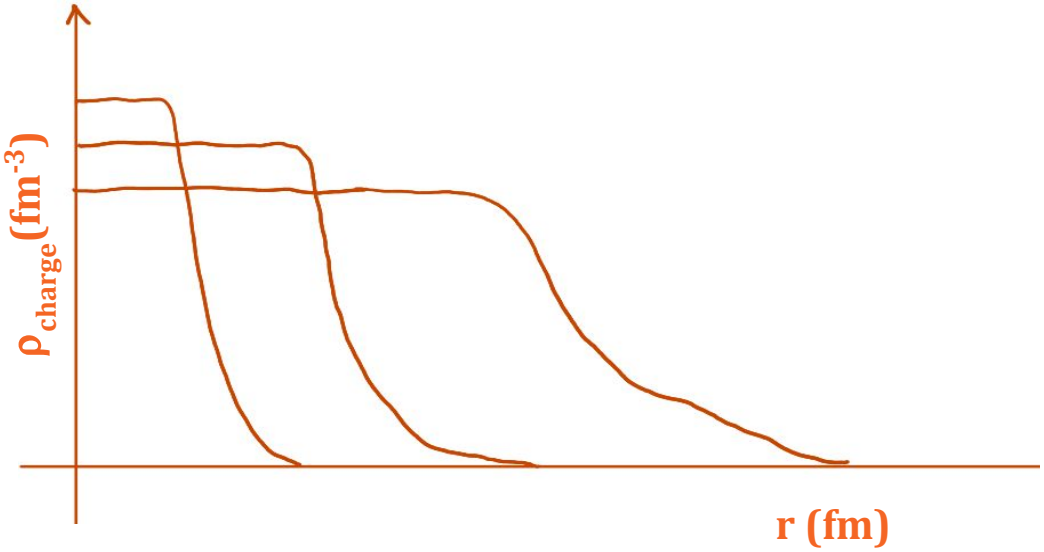


Figure 3.4 The radial charge distribution of several nuclei determined from electron scattering. The skin thickness t is shown for O, Ni, and Pb; its value is roughly constant at 2.3 fm. The central density changes very little from the lightest nuclei to the heaviest. These distributions were adapted from R. C. Barrett and D. F. Jackson, *Nuclear Sizes and Structure* (Oxford: Clarendon, 1977), which gives more detail on methods of determining $\rho(r)$.

Nuclear charge distribution (proton density)

- charge density/proton density (fm^{-3}) *vs.* distance (fm)



- density is almost constant at certain distance then it falls slowly.
- Lighter nucleus has slightly higher charge density at the center than the heavier nucleus.
- Radius at which density is almost constant is smaller for the lighter nuclei and larger for heavier nuclei.

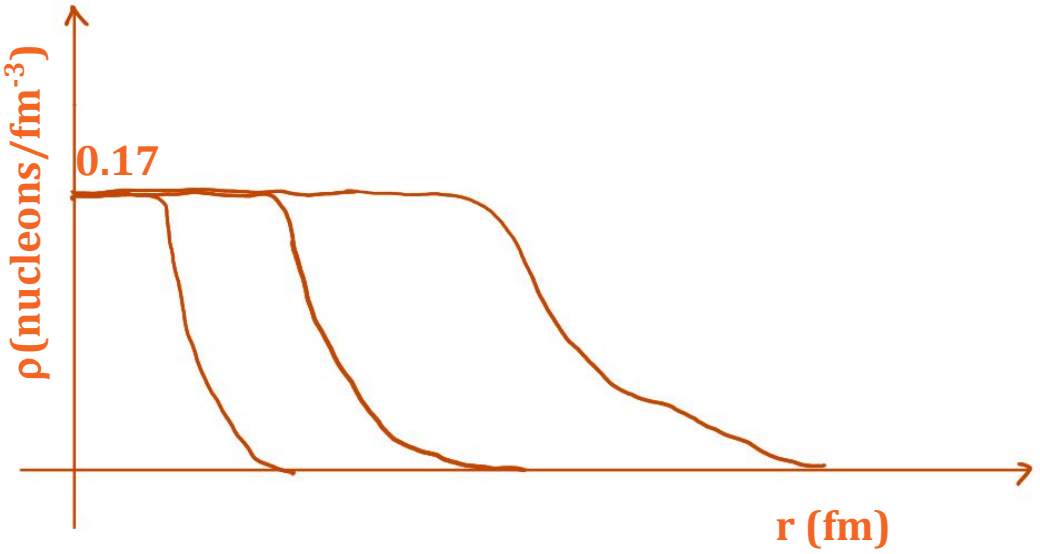
Nuclear Mass distribution (proton+neutron density)

- Let's assume that the neutrons are also similarly distributed as protons
- To obtain mass-density, we just need to scale the charge density curve:

$$\rho = \rho_{charged} \cdot \left(\frac{A}{Z}\right)$$

- A and Z are different for different nuclei.

- The mass density curve obtained after shifting the charge density curve by (A/Z) almost coincides at $\rho = 0.17$ nucleons/fm^{-3} (at r = 0)



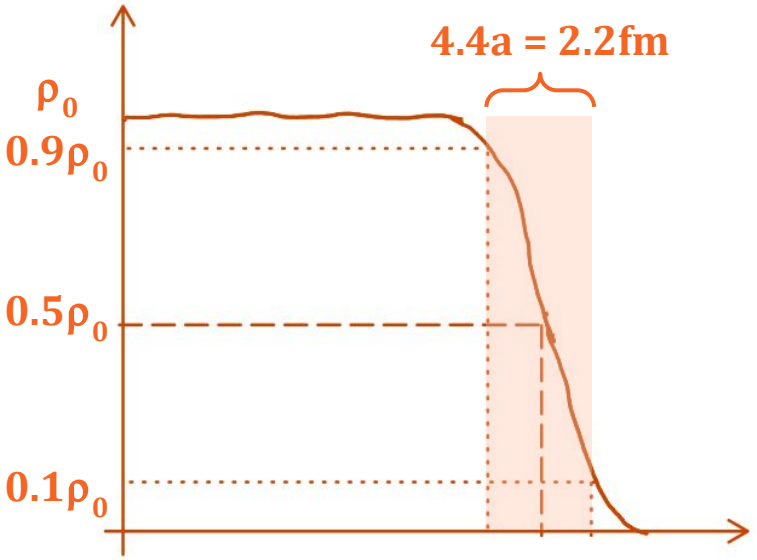
Nuclear Mass distribution (proton+neutron density)

- A particularly suitable for $\rho(r)$ is given by (fitting the experimental data):

$$\rho(r) = \frac{\rho_0}{1 + \exp\left[\frac{r-R}{a}\right]}$$

Woods-Saxon Function (Fermi distribution)

- It has three parameters:
  - ρ_0 = density at the center = 0.17 nucleons/fm^{-3} .
  - R = nuclear radius = radius at which $\rho = \rho_0/2$.
  - a = how steeply the density drops = skin thickness of the nuclei



Skin thickness of the nuclei (a)

- The thickness in which $\rho(r)$ falls from $0.9\rho_0$ to $0.1\rho_0$ at the nuclear surface = 4.4 a
- a ≃ 0.5 fm

Nuclear Mass distribution (proton+neutron density)

Let's calculate 'r' at 90% ρ_0.

$$0.9\rho_0 = \frac{\rho_0}{1+\exp\left[\frac{(r-R)}{a}\right]}$$

$$\Rightarrow \quad 1+\exp\left[\frac{(r-R)}{a}\right] = 1/0.9$$

$$\Rightarrow \quad \exp\left[\frac{(r-R)}{a}\right] = \frac{1}{0.9} - 1 = 1/9$$

$$\Rightarrow \quad \frac{r-R}{a} = \ln\left[1/9\right] = -\ln(9) = -2.1972$$

$$\Rightarrow \quad r-R = -2.1972a$$

$$\Rightarrow \quad r_{0.9} \simeq R - 2.2a$$

Similiarily, let's calculate 'r' at $0.1\rho_0$

$$\Rightarrow \quad 0.1\rho_0 = \frac{\rho_0}{1+\exp\left[\frac{(r-R)}{a}\right]}$$

$$\Rightarrow \quad 1+\exp\left[\frac{r-R}{a}\right] = 10$$

$$\Rightarrow \quad \exp\left[(r-R)/a\right] = 9$$

$$\Rightarrow \quad (r-R)/a = \ln 9 = 2.1972$$

$$\Rightarrow \quad r-R \simeq 2.2a$$

$$\Rightarrow \quad r_{0.1} \simeq R+2.2a$$

And, skin thickness $= r_{0.1} - r_{0.9}$

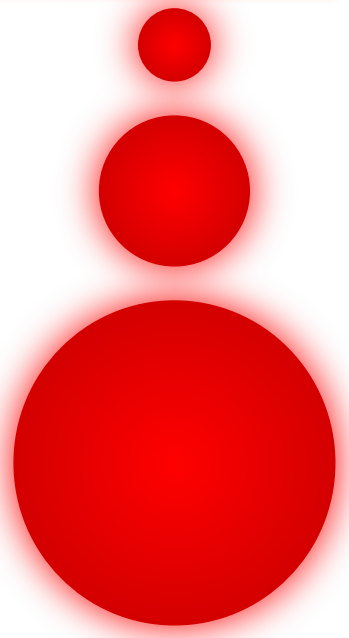
$$= R+2.2a-(R-2.2a)$$

$$= 4.4a$$

Nuclear Mass distribution (proton+neutron density)

- In almost entire range of nuclei (light to heavy):
  - a = 0.5 fm
  - ρ_0 = 0.17 nucleons/fm^{-3}

} (fitting the experimental data)



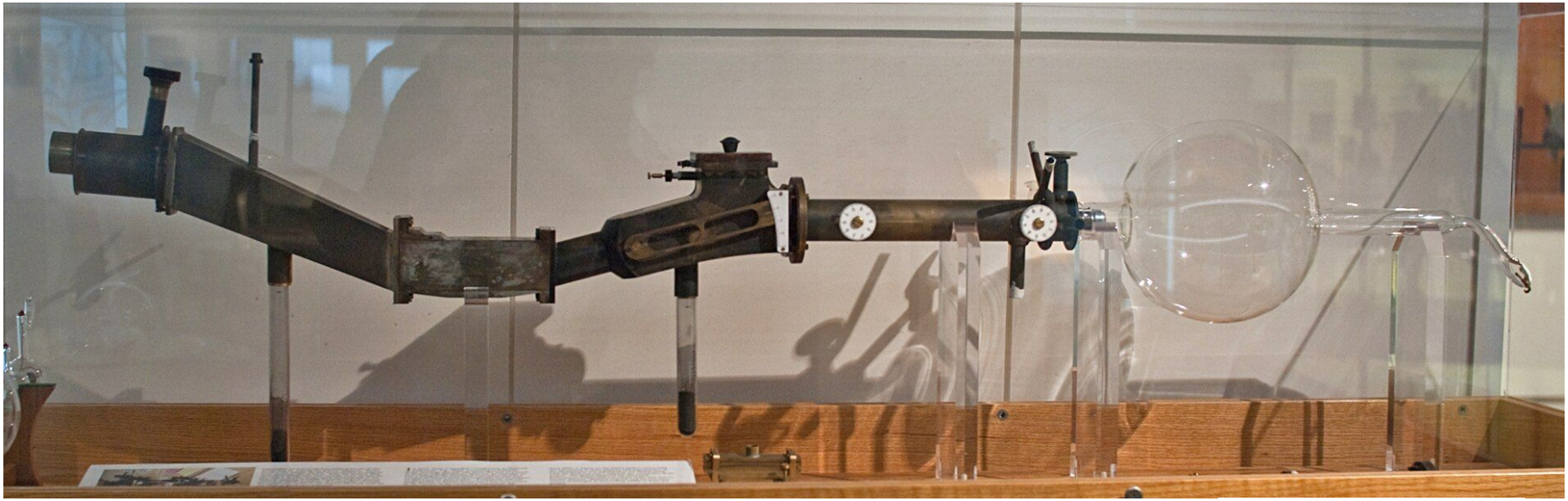
- We may assume that density of the nucleus is almost constant.
- Volume of the nuclei $\propto$ A (mass number)

  $\Rightarrow R^3 \propto A$

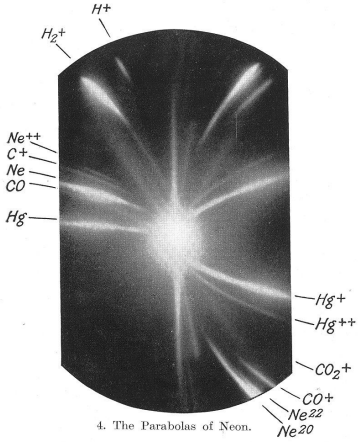
  $\Rightarrow R = R_0 A^{1/3}$
- R_0 = 1.2 fm (constant) [calculate]
- From, light to heavy nuclei the nuclear radii, R, varies as $R_0 A^{1/3}$
- However, all nuclei have same skin thickness $\simeq$ 2.2 fm

Mass Number and Isotopes

- The **atomic number** of an element is the number of protons in each of its atomic nuclei.
- Francis Aston developed the first mass spectrograph in 1919 at the Cavendish Laboratory.



- This device enabled him to successfully separate the **isotopes** ^{20}Ne and ^{22}Ne.

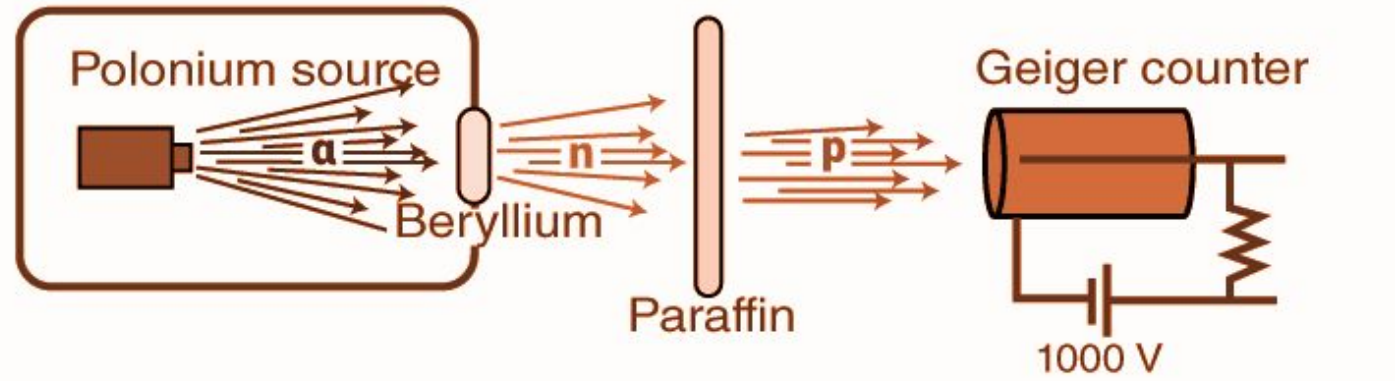


4. The Parabolas of Neon.

Neon Isotope	Relative Abundance
^{20}Ne	90.5%
^{21}Ne	0.27%
^{22}Ne	9.25%

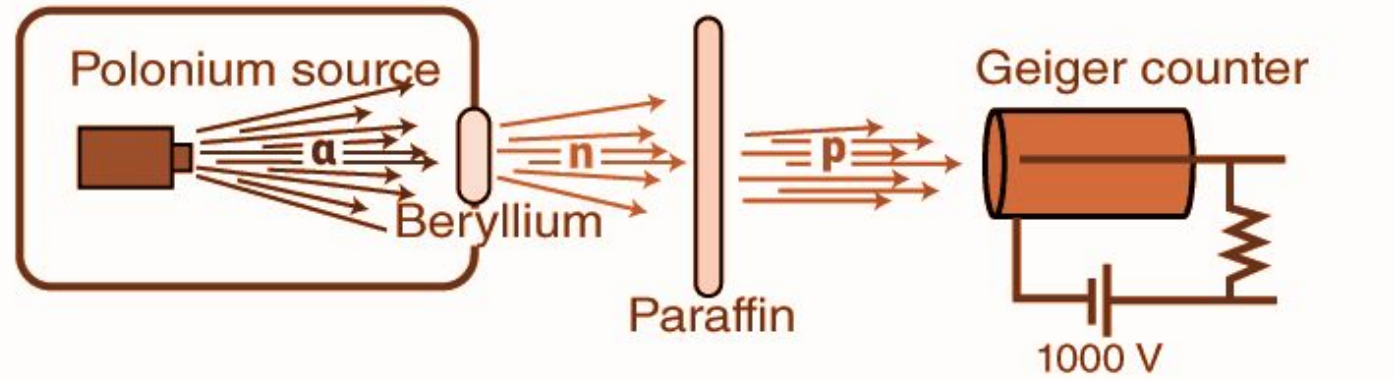
Discovery of Neutron

- It seemed there must be something in the nucleus in addition to protons.
- Nuclei of H ($\simeq 1\ m_p$), He($\simeq 4\ m_p$), Li ($\simeq 7\ m_p$), Be ($\simeq 9\ m_p$)... quite puzzling (excess mass)
- ~~Rutherford suggested (and other publications with similar suggestions) of a proton–electron composite in the nucleus ⇒ neutron.~~
- In 1930, Walther Bothe and Herbert Becker discovered a highly penetrating radiation emitted when $^{9}Be_{4}$, $^{11}B_{5}$, or $^{7}Li_{3}$ were bombarded with alpha particles from polonium.
- The newly observed penetrating radiation was unaffected by electric fields, leading scientists to believe it was gamma radiation.
- In 1932, Irène Joliot-Curie and Frédéric Joliot discovered that the unknown radiation ejected high-energy protons (5 MeV) when it struck paraffin wax.
- Based on energy and momentum calculations, a gamma ray would require an unrealistically high energy level (50 MeV) to eject a massive proton.



Discovery of Neutron

- Chadwick replicated the radiation by bombarding beryllium with alpha particles
- Analyzing the recoil energy of the protons revealed that the radiation particles had a mass comparable to that of a proton, disproving the gamma ray theory.
- Uncharged particles with a mass similar to protons exhibited the characteristics described by Rutherford in 1920, later identified as neutrons.
- Chadwick won the Nobel Prize in Physics in 1935 for this discovery.



Nuclear Mass and Binding Energy

- The proton and the neutrons are held together inside the nucleus by very short range attractive force.
- The sum of the numbers of the neutrons (N) and protons (Z) inside the nucleus is known as the mass number A:

$$A = N + Z$$

- Nuclei with same Z (protons), but different A are called isotopes ⇒ the nuclei contains same number of protons but different number of neutrons.
- Relative abundance is the percentage of atoms with a specific atomic mass found in a sample of an element.

	abundance	half-life ($t_{1/2}$)
^{1}H	99.9855%	stable
^{2}H	0.0145%	stable
^{3}H	trace	12.32 y

	abundance	half-life ($t_{1/2}$)
^{3}He	0.0002%	stable
^{4}He	99.9998%	stable

^{63}Cu	69.2%	stable
^{64}Cu	synth	12.70 h
^{65}Cu	30.9%	stable
^{67}Cu	synth	61.83 h

^{194}Hg	synth	444 y
^{195}Hg	synth	9.9 h
^{196}Hg	0.15%	stable
^{197}Hg	synth	64.14 h
^{198}Hg	10.0%	stable
^{199}Hg	16.9%	stable
^{200}Hg	23.1%	stable
^{201}Hg	13.2%	stable
^{202}Hg	29.7%	stable
^{203}Hg	synth	46.612 d
^{204}Hg	6.82%	stable

- Nuclei with the same A, but different Z are known as isobars.
- Nuclei with the same number of neutrons are known as isotones.

	abun-dance	half-life ($t_{1/2}$)
^{105}Ag	synth	41.3 d
^{106m}Ag	synth	8.28 d
^{107}Ag	51.8%	stable
^{108m}Ag	synth	439 y
^{109}Ag	48.2%	stable
$^{110m2}Ag$	synth	249.86 d
^{111}Ag	synth	7.43 d

Nuclear Mass and Binding Energy

- We need to subtract the mass of electrons from the experimental measurements of atomics using mass spectroscopy techniques to get the 'bare' nuclear mass:

$$M_{nuc} = M - Z.m_e$$

(in the above formula the B.E. of atomic electrons are neglected, which are anyways a small factor)

- **Binding energy of nucleus** : The mass of a nucleus is less than the combined mass of its individual protons and neutrons. This discrepancy between the expected and actual nuclear mass is Nuclear Binding Energy .

- A nucleus formation releases energy equivalent to its binding energy.

$$E_B = \Delta M c^2 = \{Z.M_H + N.M_n - M(A,Z)\}c^2$$

M(A,Z) : Mass of the "atom"
M_H : Mass of the Hydrogen atom.
Mn : Mass of the neutron
(notice that mass of the electron is actually cancelling out)

Nuclear Mass and Binding Energy

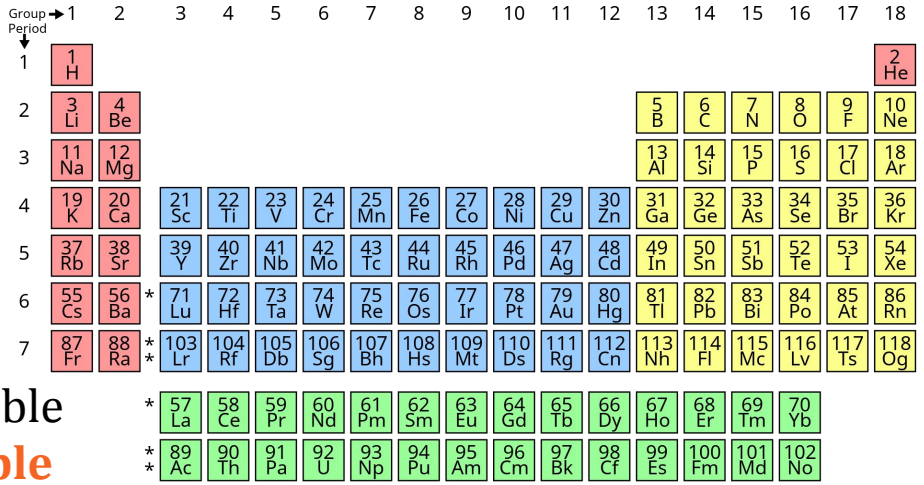
- The unit of atomic mass is defined to be 1/12th of the mass of the ^{12}C atom taken to be exactly 12 units and is designated by the symbol u (unified atomic mass unit).
- $m_u c^2 = 1.49241808768(46)\times10^{-10}$ J = 931.49410372(29) MeV

Particle	Mass (kg)	Mass (u)	Mass (MeV/c^2)
electron	$9.1093837139(28)\times10^{-31}$	$5.485799090441(97)\times10^{-4}$	0.51099895069(16)
proton	$1.67262192595(52)\times10^{-27}$	1.007276466789(83)	938.27208943(29)
neutron	$1.67492749804(95)\times10^{-27}$	1.00866491588(49)	939.56542052(54)

- Atomic masses can be determined with accuracies better than one part in a million by modern mass spectroscopy.
- Such great accuracies are needed for the determination of nuclear binding energies and in the calculation of nuclear disintegration energies.

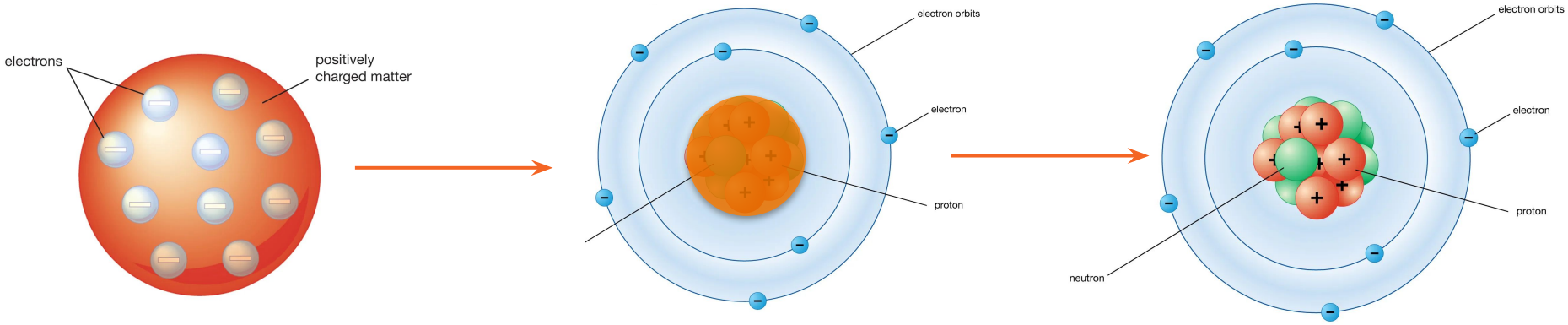
Particle Physics : Quest for Fundamental

- Dalton's **Atomic Theory** (~1800):
  - All matter is made of very tiny particles called atoms.
  - Atoms are indivisible particles, which cannot be created or destroyed in a chemical reaction.



- Grouped atoms based on their comparable chemical characteristics ⇒ **periodic table**

But is the atom fundamental?



(after Rutherford's experiment)

The diagrams are schematic, strictly NOT to scale.

(after the discovery of the proton and the neutron)

- This is just a **beginning of the quest for 'Fundamentals'**.

Particle Physics : Quest for Fundamental (fast forward… ⋙)

- 1897 - discovery of electron
- 1920 - discovery of proton
- 1932 - discovery of neutron
- 1932 - discovery of positron - antimatter!!!

- Relativistic energy relation:

$$\Rightarrow E^2 = p^2c^2 + m^2c^4$$

$$\Rightarrow E = \pm\sqrt{(p^2c^2 + m^2c^4)}$$

- For every positive-energy solution $E = \sqrt{(p^2c^2 + m^2c^4)}$ there is a corresponding solution with negative energy $E = -\sqrt{(p^2c^2 + m^2c^4)}$
- Then, why not electrons just populate the negative energy states.

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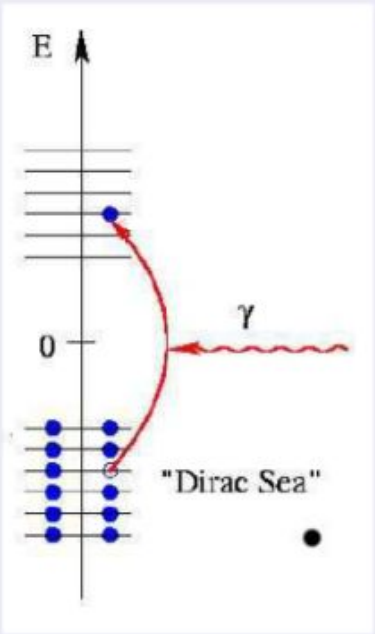
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Anti-particles

- Dirac's proposal for solutions with $E < 0$: Fill the (Dirac-)"sea" of negative energy states, (Fermi-character prevents double fillings and therefore guarantees the stability of the vacuum).
- Can excite them with, e.g., photons.
- Then anti-particles are just "holes" in the sea: **absence of negative energy looks like net positive energy**.
- The related particle must have **same mass** as ordinary particles, but **opposite charge**.

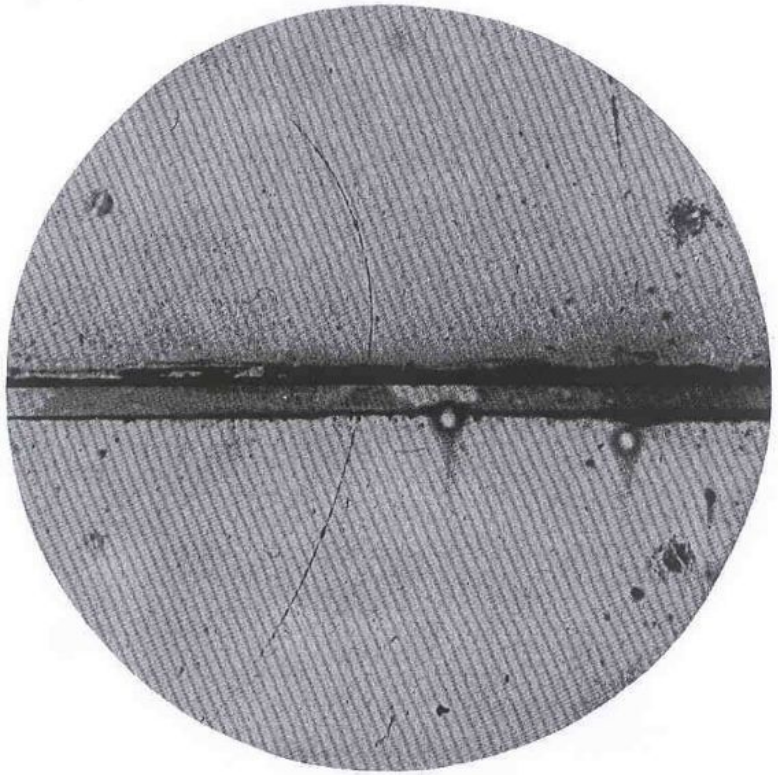


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- Carl D. Anderson (Caltech) [1932] found evidence for an electron with a positive charge, or positron. Anderson discovered the positron while using a cloud chamber to investigate cosmic rays.



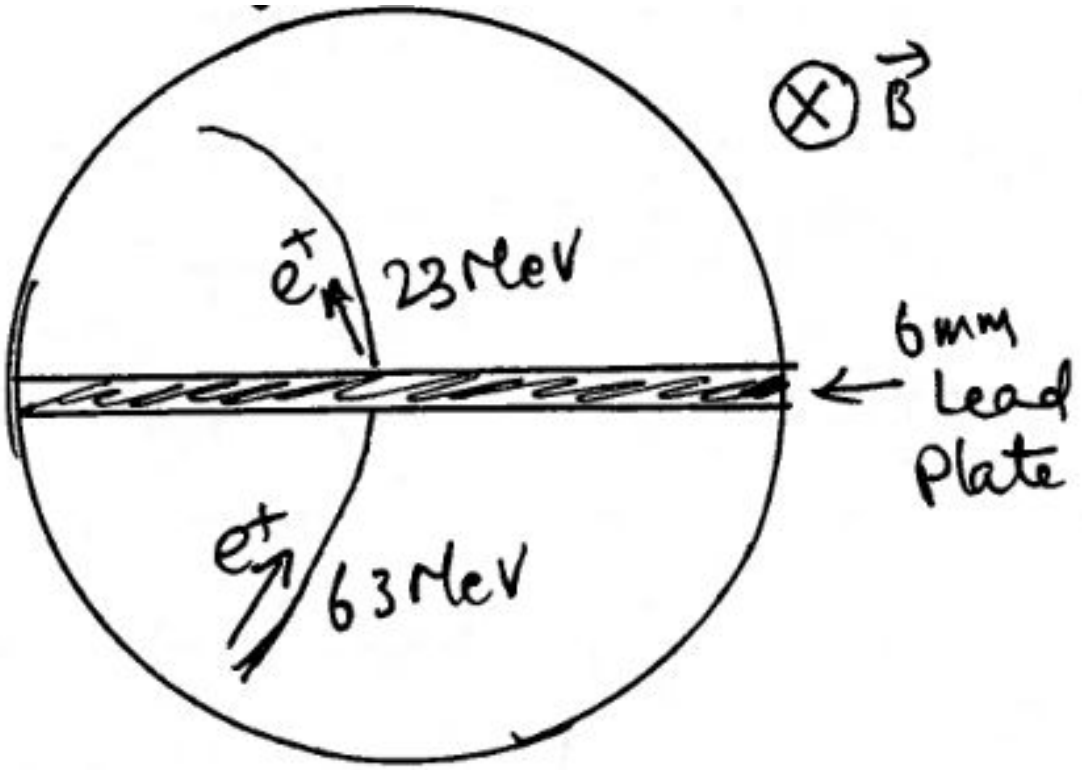


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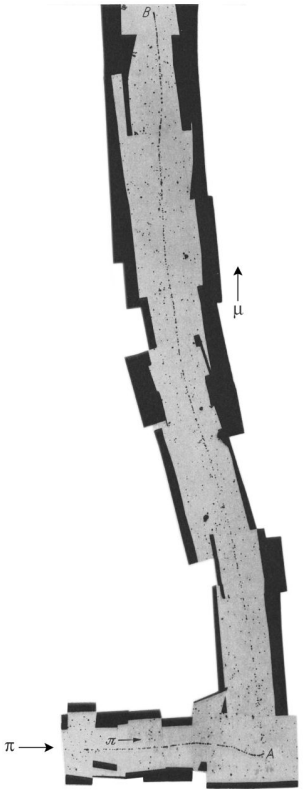




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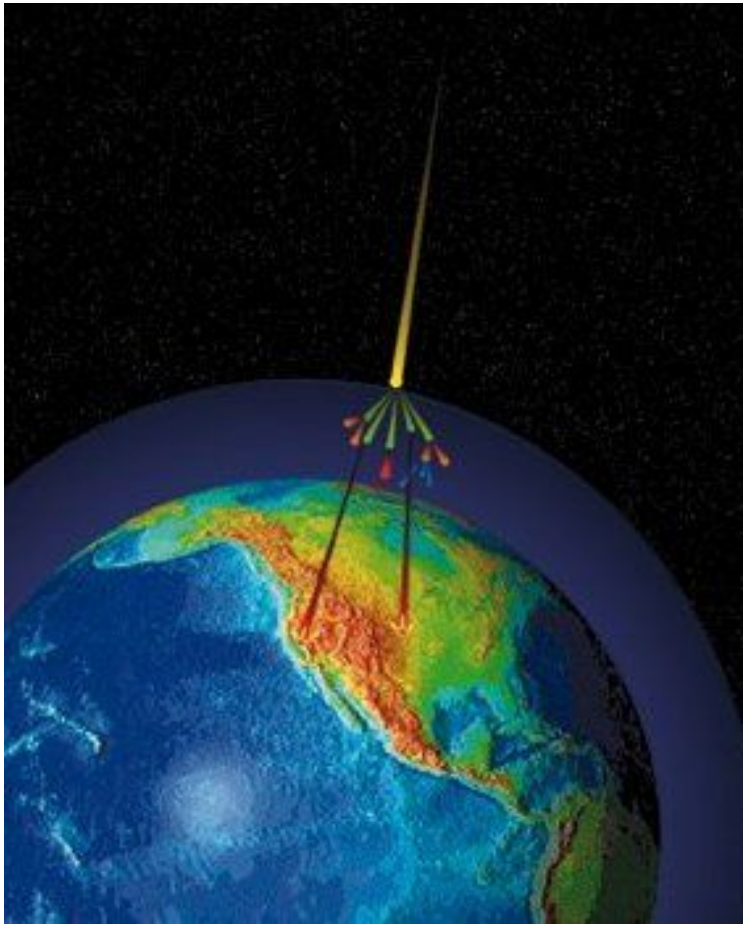
  - The eventual recognition of the muon as a simple "heavy electron", with no role at all in the nuclear interaction, seemed so incongruous and surprising at the time, that Nobel laureate I. I. Rabi famously quipped, "Who ordered that?"

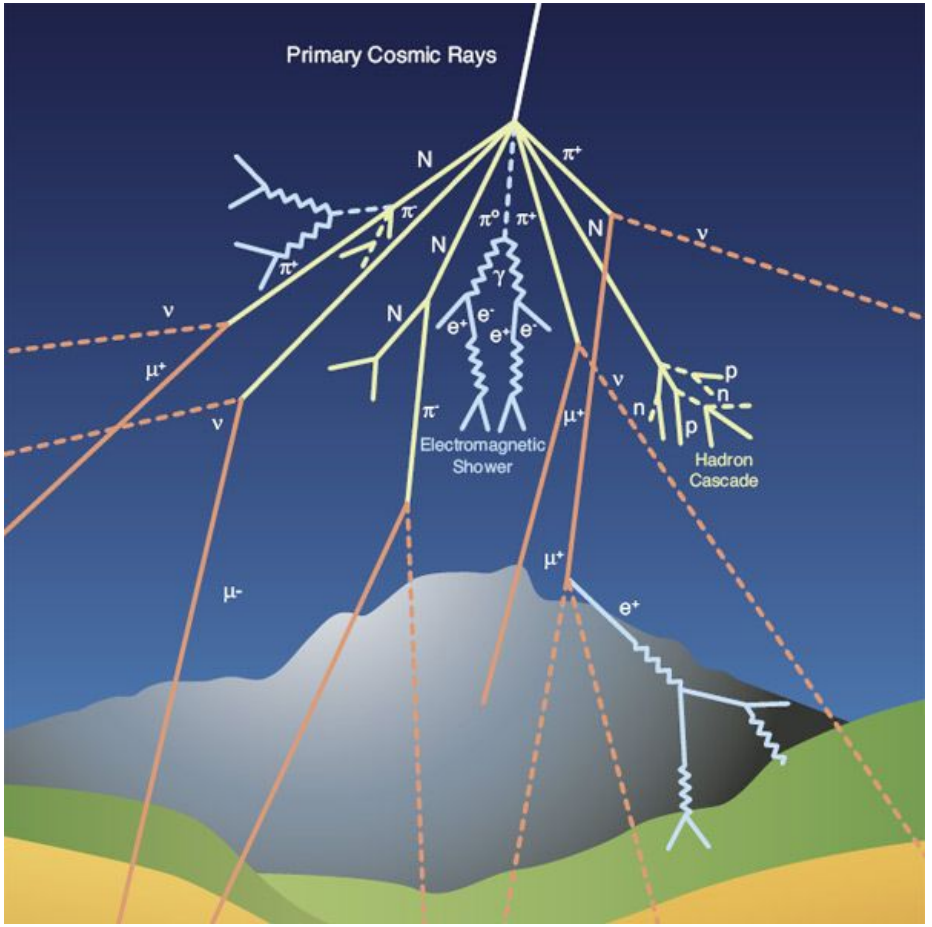


  - Powell's earliest pictures showing the track of a pion in a photographic emulsion exposed to cosmic rays at high altitude. The pion (entering from the left) decays into a muon and a neutrino (the latter is electrically neutral, and leaves no track)

Cosmic Ray and Muons

- Cosmic rays are a form of high-energy radiation that originate from outside our solar system.
- When they reach Earth, the rays collide with particles in the upper atmosphere to produce a "shower" of particles, including muons.



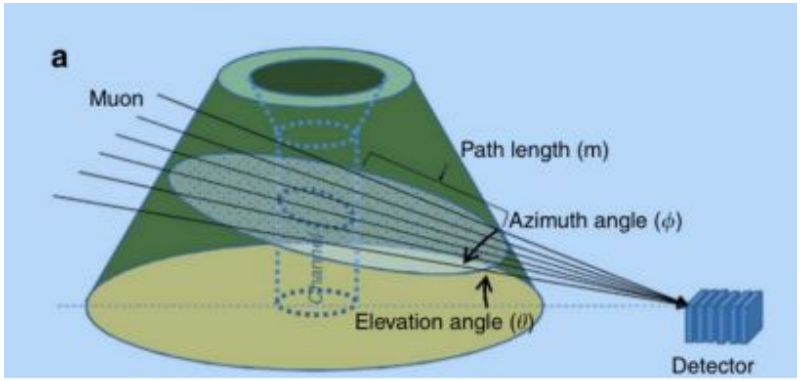


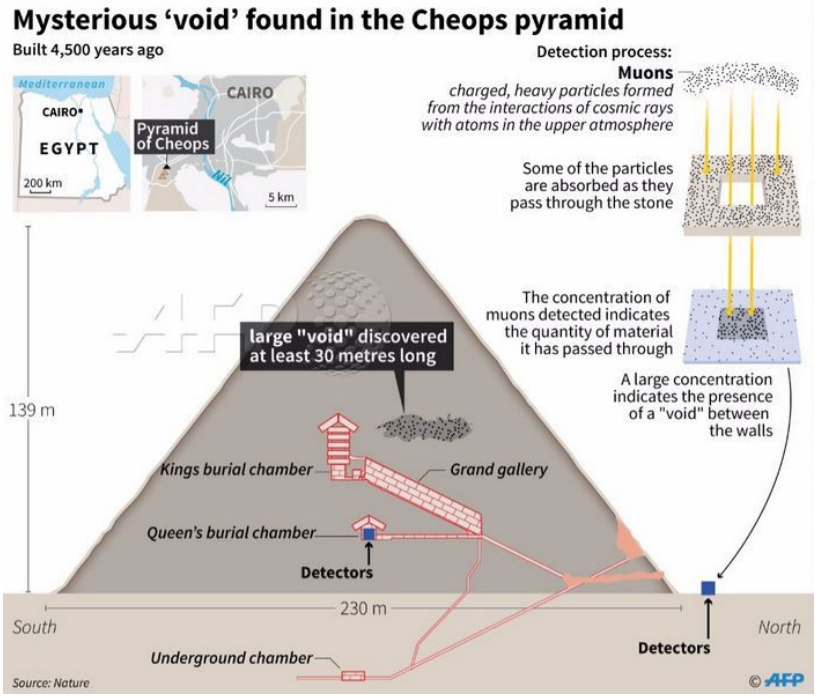
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- Approximately one muon hits every square centimeter of the Earth every minute at sea level. This rate of natural background radiation increases at higher elevations.

Applications:

- The ScanPyramids mission found a big void in the Great Pyramid (Khufu's Pyramid), above the Grand Gallery (Nature, 2017).



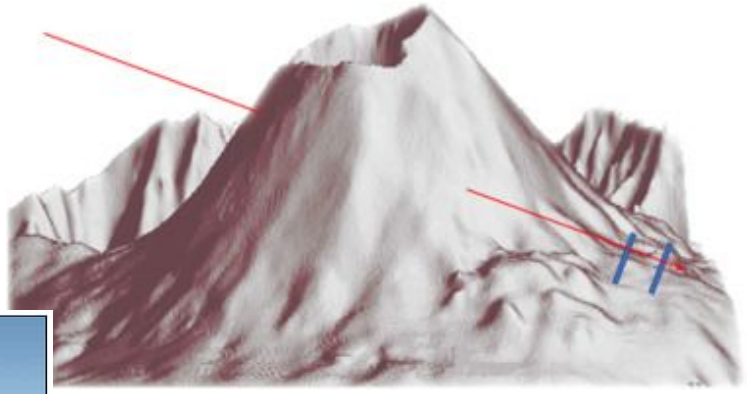


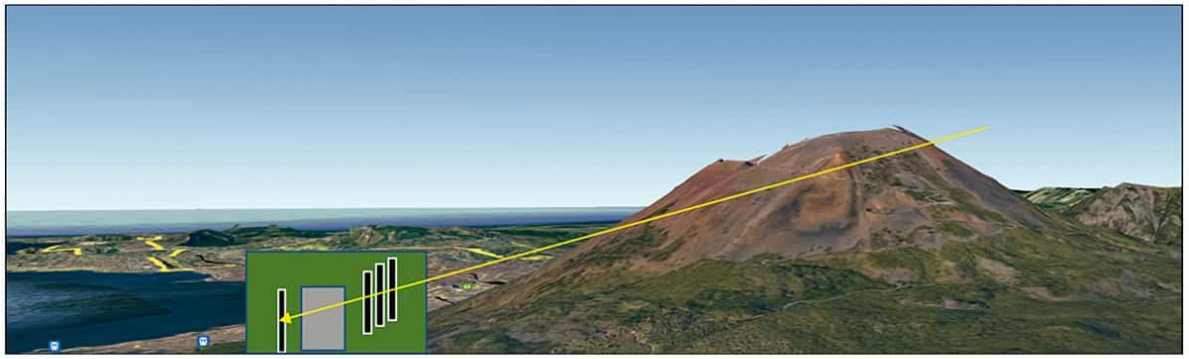
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Applications:

- Geophysical radiography techniques using cosmic muon particles have been proposed to study the detailed understanding of a volcano inner structure.
  - The MURAVES experiment, based at Mount Vesuvius in Italy, aims to capture muographic images of the internal structure of the summit of the volcano.



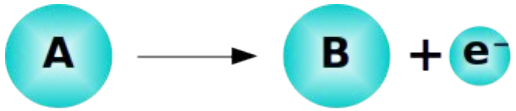


Particle Physics : Quest for Fundamental (fast forward… ⋙)

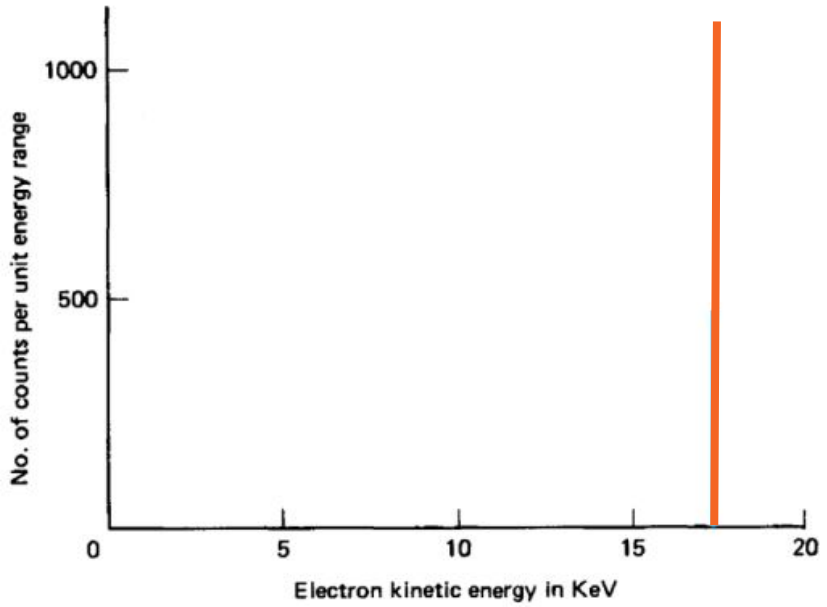
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- 1956 - **discovery of electron-neutrino**

- Proposed as a missing energy (by Pauli) in nuclear β-decays.

- β decay (before discovery of neutrino): In the frame parent nucleus (A) is at rest , the nucleus (B) and e^- comes out back to back with equal and opposite momenta:



Expectation⇒

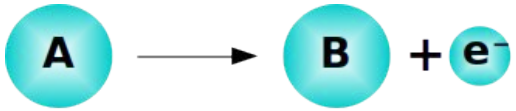


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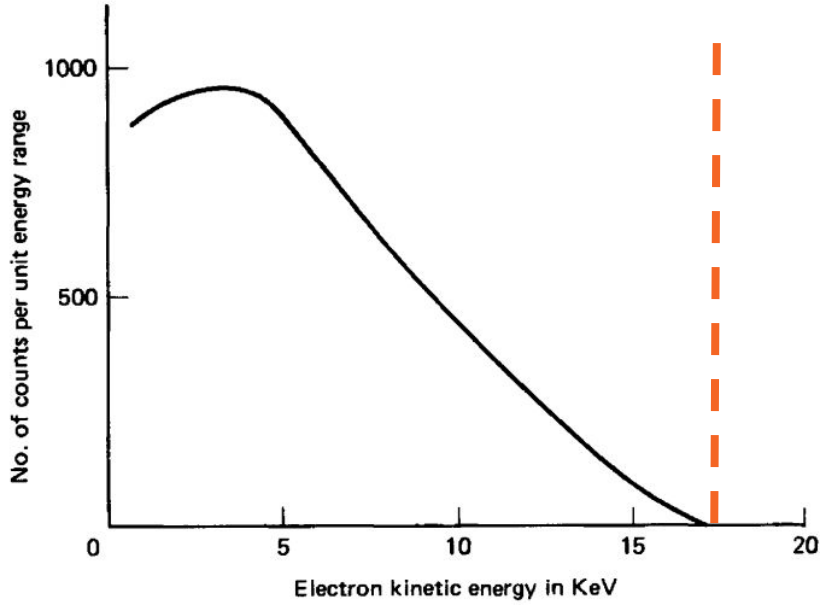
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Reality⇒

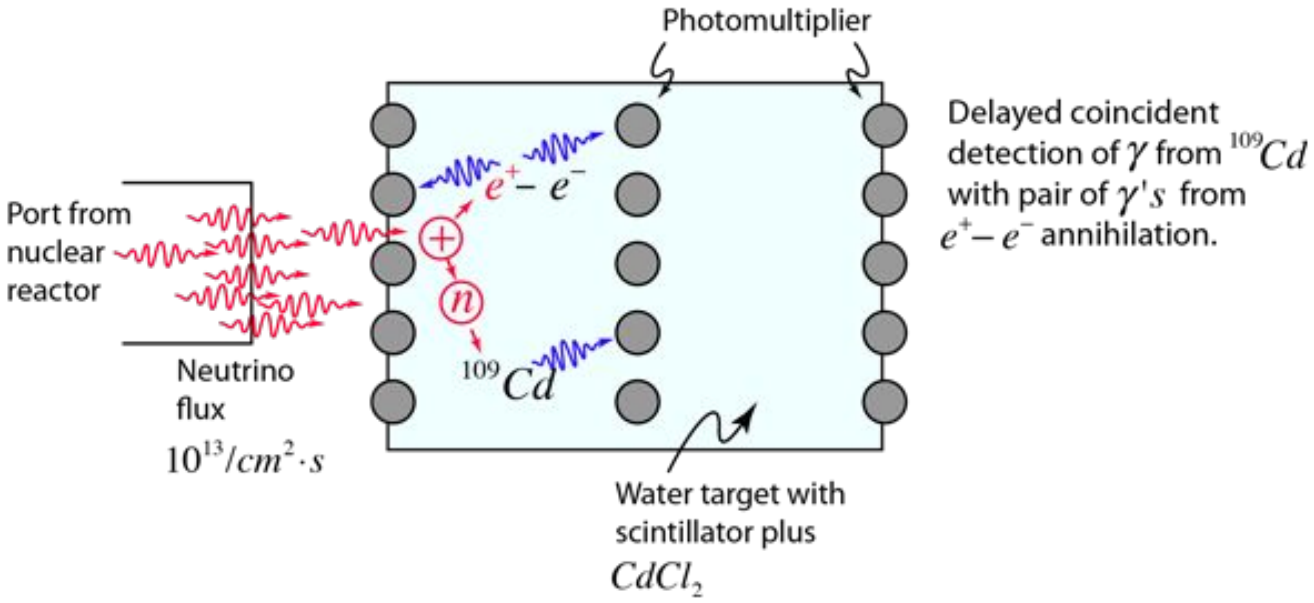


Discovery of electron neutrino

- This experimental result of β-decay energy spectrum was a disturbing result. “Niels Bohr was ready to abandon the law of conservation of energy.”
- Pauli suggested that another particle was emitted along with the electron, that carries off the “missing” energy. (Pauli proposed to call it the neutron)
- From the fact that the observed β spectrum ranges up to the value obtained from the 2-body calculations : the proposed particle has tiny mass.
- Fermi called it the neutrino (anti-neutrino).
- By 1950, there was compelling theoretical evidence for the existence of neutrinos, but there was still no direct experimental verification.
- Why? Detection of neutrino is so difficult?
- The decisive experiments were conducted at the Savannah River nuclear reactor in South Carolina, in the mid-fifties.
- Cowan and Reines set up a large tank of water ($CdCl_2$ dissolved) and watched for the “inverse” β-decay reaction.
- The reactor had a neutrino flux of 5×10^{13} neutrinos per second per square centimeter

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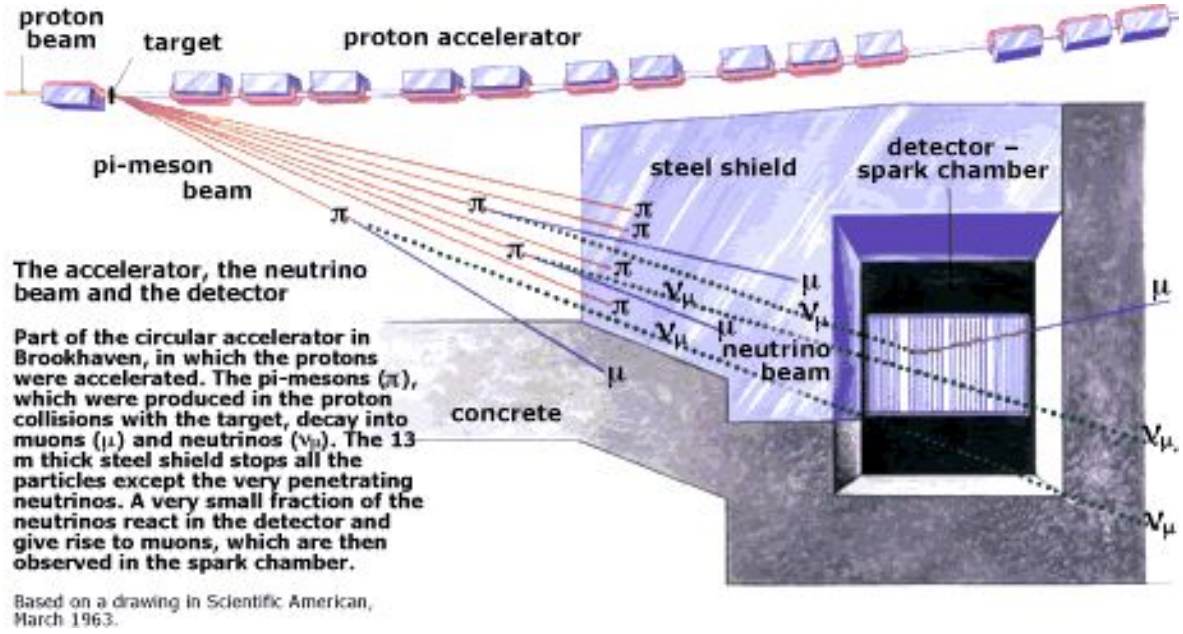
(million dollar particle)

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- 1962 - discovery of muon-neutrino

Table top experiments

Need of accelerators
Need of Large Scale Detectors





Leptons

Generation:	I	II	III
charged:	≈0.511 MeV/c², −1, ½, e, electron	≈105.66 MeV/c², −1, ½, μ, muon	≈1.77693 GeV/c², −1, ½, τ, tau
neutral:	<0.8 eV/c², 0, ½, ν_e, electron neutrino	<0.17 MeV/c², 0, ½, ν_μ, muon neutrino	<18.2 MeV/c², 0, ½, ν_τ, tau neutrino

- Leptons do not participate in the strong interaction.
- Charged leptons (electron, muon and tau) participate in the “weak” as well as “electromagnetic interaction”.
- Neutral leptons (electron-neutrino, muon-neutrino and tau-neutrino) participate only in the “weak” interaction.
- They have their corresponding anti-particles:
  - electron: positron; muon : anti-muon; tau : anti-tau
  - electron-neutrino : anti-electron-neutrino; muon-neutrino : anti-muon-neutrino ; tau-neutrino : anti-tau-neutrino

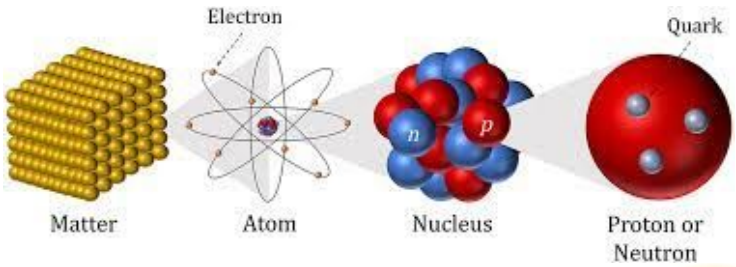
Particle Physics : Quest for Fundamental (fast forward… »)

- 1897 - discovery of electron
- 1920 - discovery of proton
- 1932 - discovery of neutron
- 1932 - discovery of positron - antimatter.
- 1936 - discovery of muons (heavy version of electron)
- 1956 - discovery of electron-neutrino
- 1962 - discovery of muon-neutrino
- 1968 - discovery of (light) quarks

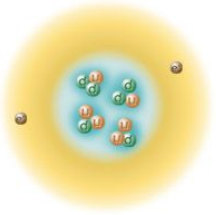
Table top experiments

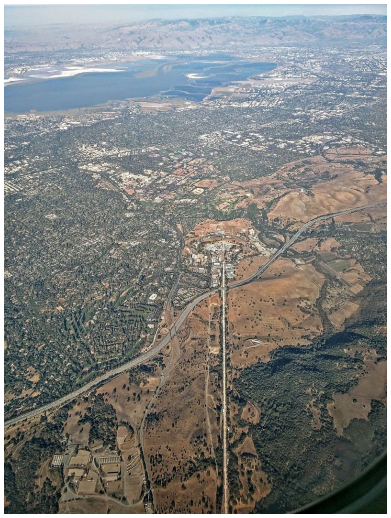
Need of accelerators
Need of Large Scale Detectors

- Large sized scattering experiment designed.
- bombarded H-targets using high-energy e- beam (20 GeV) at SLAC.
- Why high energy beams needed?



Current picture of atom (till date) →



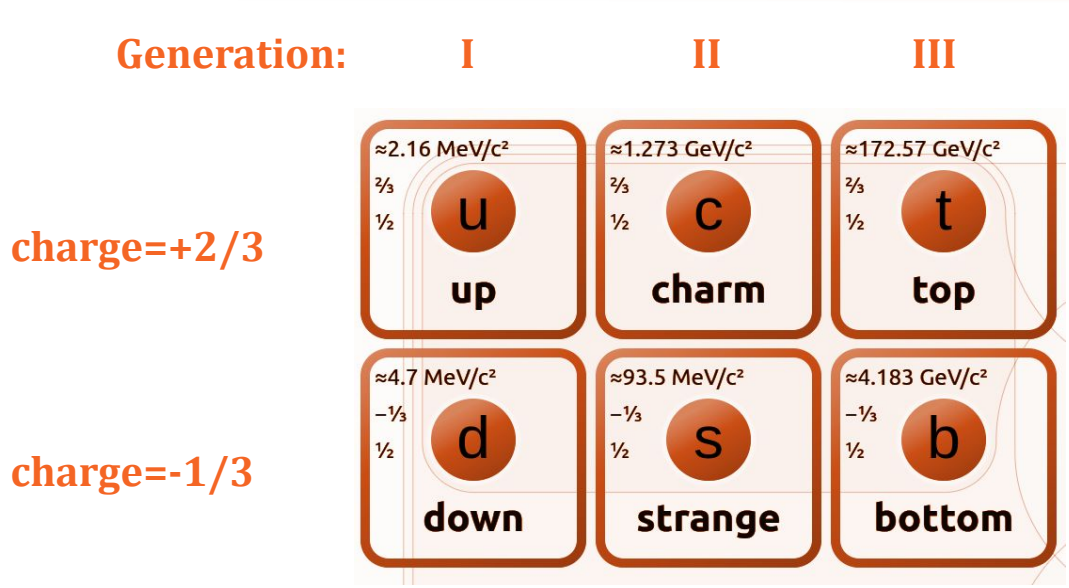


A 3.2 km long electron accelerator energy upto 50 GeV.

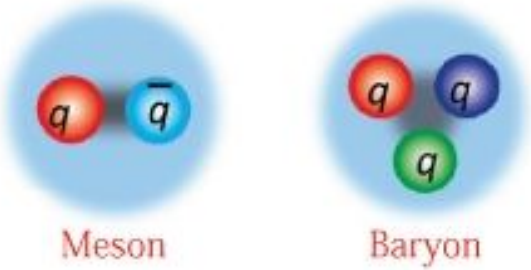


SLAC detectors that helped establish the existence of quarks

Particle Physics : Quest for Fundamental (fast forward… »)



- Quarks participate in the strong, weak and electromagnetic interaction.
- But no one has seen any free quark : "So quarks ought to be easy to produce, easy to identify, and easy to store, and yet, no one has ever found one."
- quark confinement: quarks are absolutely confined within baryons and mesons

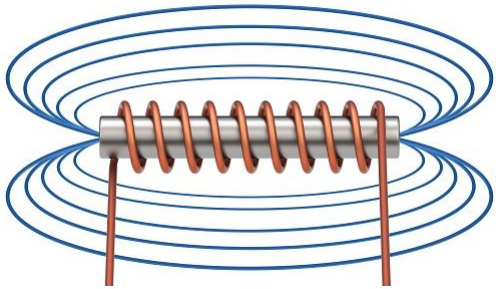


Fundamental Forces/interaction

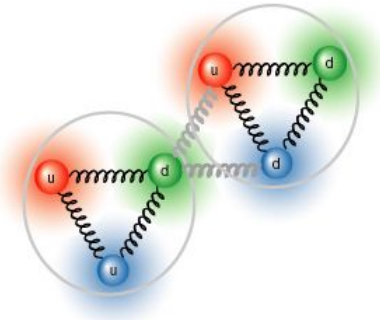
Gravity:



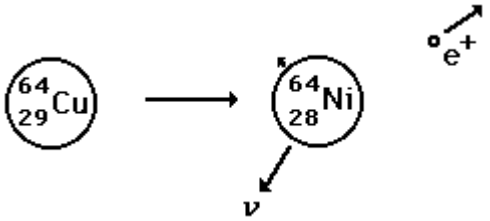
Electromagnetic:



Strong:



Weak:

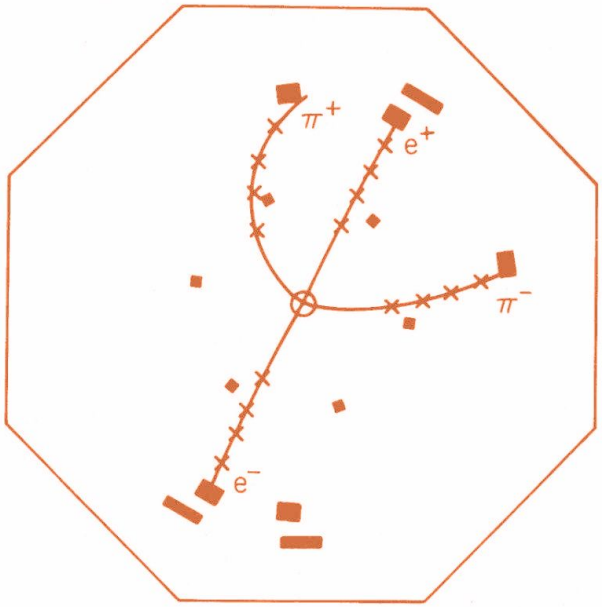


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- 1962 - discovery of muon-neutrino
- 1968 - discovery of (light) quarks
- 1974 - discovery of charm quark
- 1975 - discovery of tau-lepton
- 1977 - discovery of bottom quark
- 1979 - discovery of gluons
- 1983 - discovery of W and Z bosons
- 1995 - discovery of top quark
- 2000 - discovery of tau neutron
- 2012 - discovery of Higgs boson

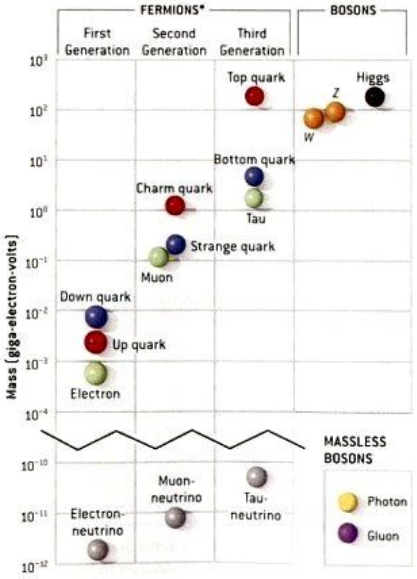
Table top experiments

Need of accelerators
Need of Large Scale Detectors

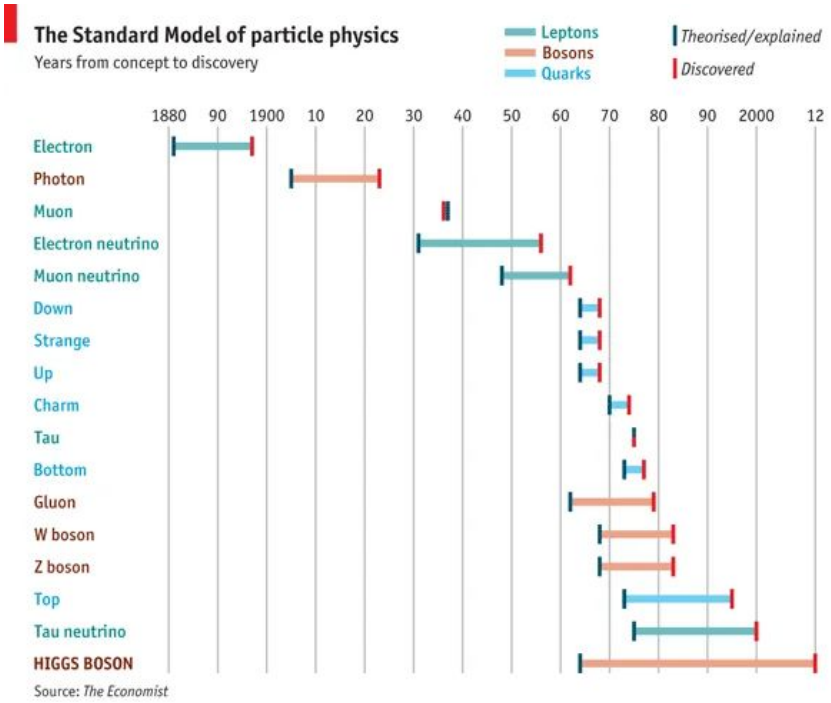


50 years of "november revolution" in particle physics

Particle Physics : Need of High Energy (Mega) experiments



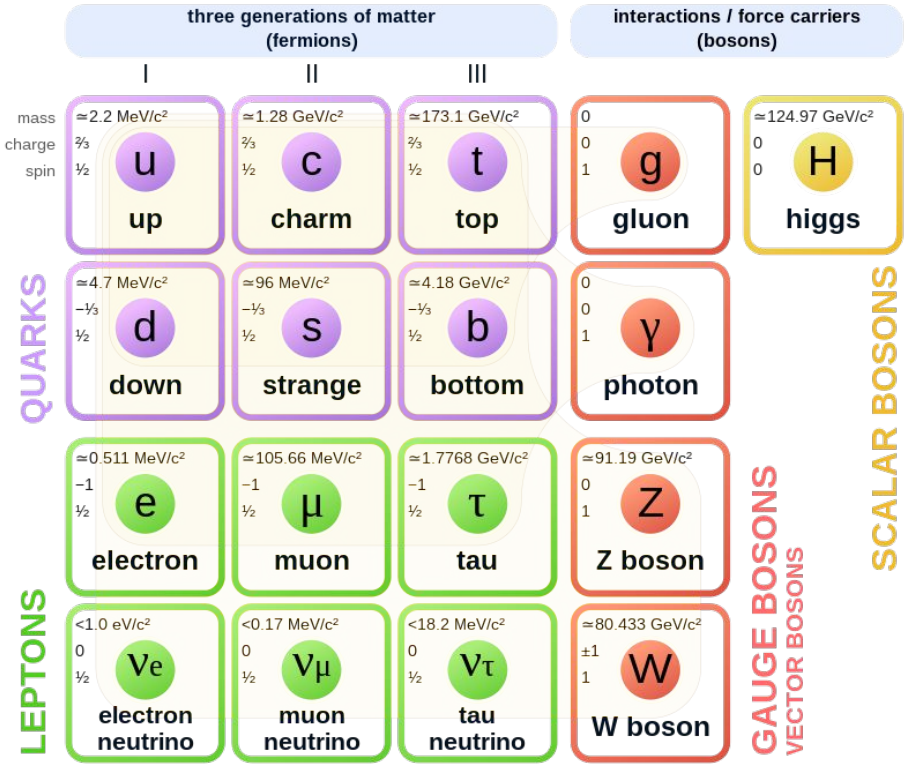
Particle Mass increasing (roughly)



Need of Higher Energy :

- Finer Spatial resolution of the 'probe' (beam) used to investigate possible (internal) structure
  - the resolution $\Delta r \propto \lambda \propto 1/p$ or, $1/E$
  - Higher the energy $\Rightarrow$ smaller (better) the resolution $\Rightarrow$ can probe smaller internal structure
- Elementary particles (higher gen.) are massive and the energy (mc^2) required to produce them in laboratory is very high. [Mass of top quark ~ 173 GeV].
- Beyond SM particles could be much heavier and the searches are still on.

The Standard Model of Particle Physics



Are we done? No!

- Normal Matter is less than 5%, What is the origin and genesis of dark matter/energy.
- The cosmological antimatter mystery.
- Parameters of Standard Model [Einstein: "there are no arbitrary constants …"]

More unresolved questions..



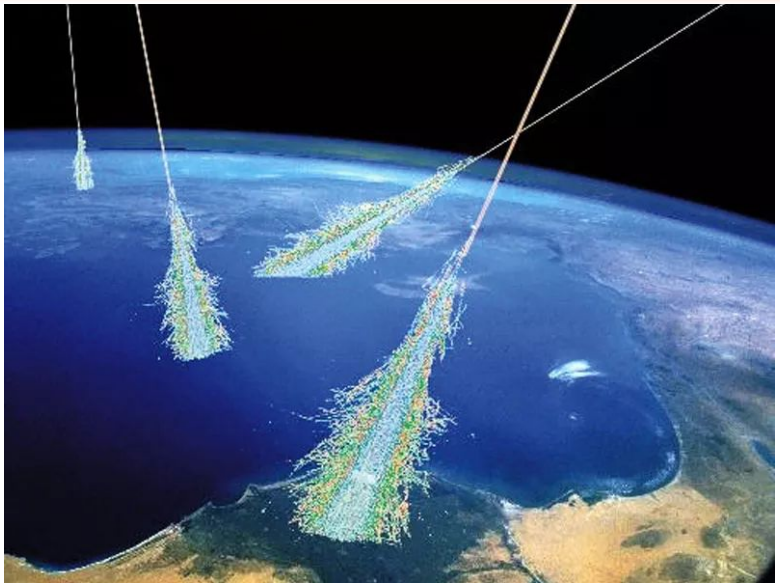
Not a complete list …

- 1935 - neutron
- 1936 - positron
- 1949 - nuclear forces
- 1957 - parity-violation
- 1959 - antiproton
- 1960 - bubble chamber
- 1965 -quantum electro-dynamics
- 1969 - eight-fold way (→quark model)
- 1976 - J/ψ (charm quark)
- 1979 - Electro-weak unification theory
- 1980 - CP-violation in Kaon-sector
- 1984 - W and Z Bosons
- 1988 - muon-neutrino
- 1990 - DIS, quarks
- 1992 - multiwire proportional chamber
- 1995 - tau lepton and neutrino
- 1999 - Quantum Field Theory
- 2004 - Asymptotic Freedom
- 2008 - CP violation, 3 gen of quarks and spontaneous broken symmetry.
- 2013 - Higgs Boson
- 2015 - Neutrino oscillation

How to produce elementary particles?

Cosmic Rays:

- The earth is constantly bombarded with high-energy particles – mostly protons, some are He-nuclie and rest are heavy nuclei.
- Hit upper atmosphere and produce shower of secondary particles.
- Advantage : free, energy can be enormous.
- Disadvantage: The rate is low, uncontrollable requires patience.





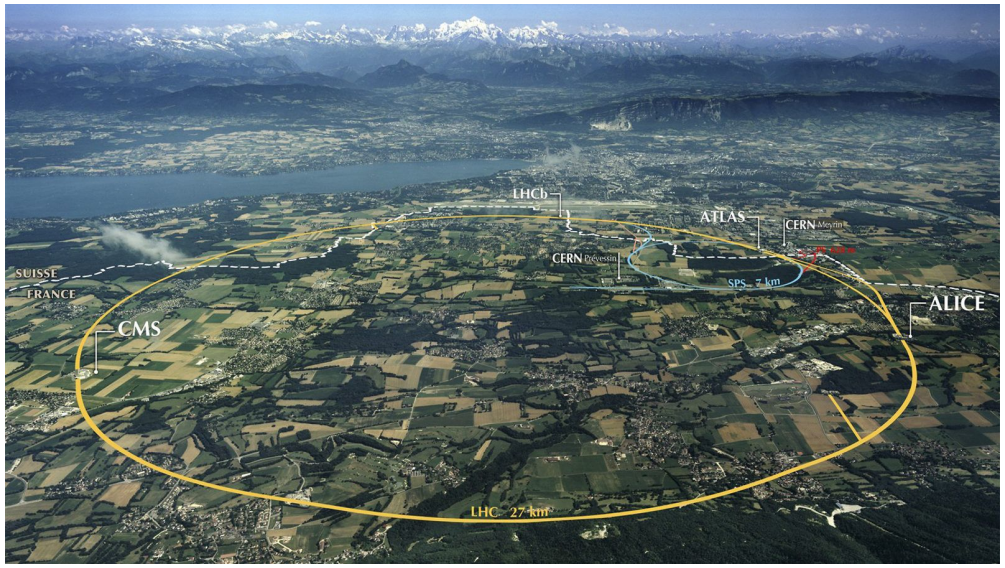
Nuclear Reactors:

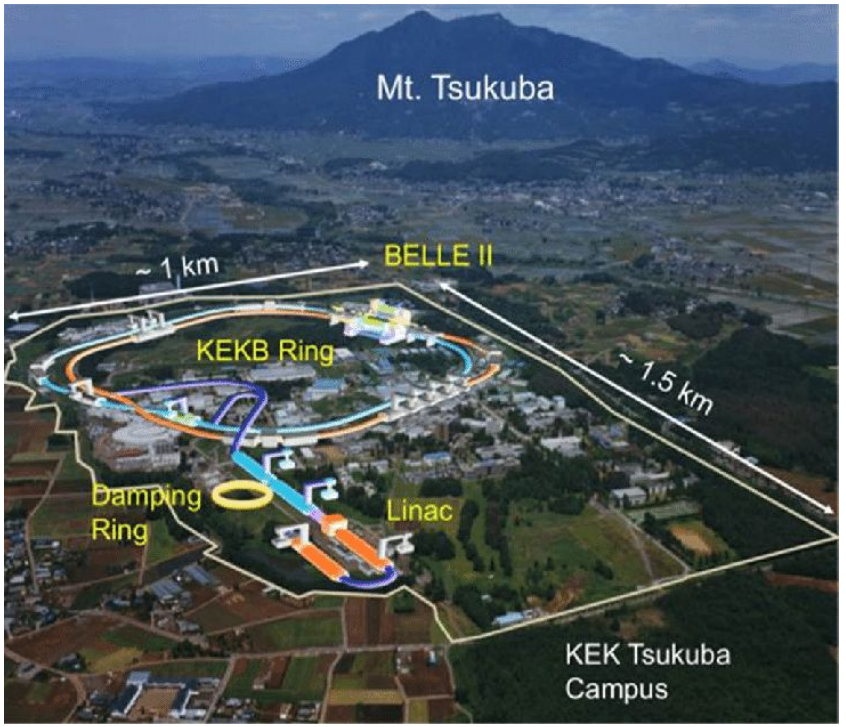
- Fission inside of the nuclear reactor eventually leads to the birth of neutrinos through beta decay.
- Important place to study 'neutrino physics'

How to produce elementary particles?

Accelerators:

- Accelerate the particles at high energies → collide them to a fixed target/opposite beam.
- Study the collision debris, by putting detectors around the collision points.
- Higher the mass of the particle to be produced, higher should be the energy of the colliding particles.
- Few examples: LHC at CERN, SuperKEKB at KEK ..







భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Thank You!